

# US Patent & Trademark Office

## Patent Public Search | Text View

United States Patent Application Publication

20250284086

Kind Code

A1

Publication Date

September 11, 2025

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### FOLDED CAMERA LENS DESIGNS

#### Abstract

Digital cameras, optical lens modules for such digital cameras and methods for assembling lens elements in such lens modules. In various embodiments, the digital cameras comprise an optical lens module including  $N \geq 3$  lens elements  $L_{sub.i}$ , each lens element comprising a respective front surface  $S_{sub.2i-1}$  and a respective rear surface  $S_{sub.2i}$ . In various embodiments the first lens element toward the object side,  $L_{sub.1}$  and its respective front surfaces  $S_{sub.1}$  have optical and/or mechanical properties, such as a clear aperture, a clear height and a mechanical height that are larger than respective properties of following lens elements and surfaces. This is done to achieve a camera with large aperture stop, given a lens and/or camera height.

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**Family ID:** 63252479

**Appl. No.:** 19/218688

**Filed:** May 27, 2025

#### Related U.S. Application Data

parent US continuation 18314866 20230510 parent-grant-document US 12345943 child US 19218688

parent US continuation 17665495 20220205 parent-grant-document US 11668894 child US 18314866

parent US continuation 16851475 20200417 parent-grant-document US 11347020 child US 17665495

parent US continuation 16392754 20190424 parent-grant-document US 10670827 child US 16851475

parent US continuation 16252608 20190119 parent-grant-document US 10571644 child US 16392754

parent US continuation 16077477 20180812 parent-grant-document US 11347016 WO  
continuation PCT/IB2018/050988 20180218 child US 16252608  
us-provisional-application US 62462438 20170223  
us-provisional-application US 62478783 20170330  
us-provisional-application US 62483422 20170409

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## Publication Classification

**Int. Cl.:** **G02B7/02** (20210101); **G02B7/08** (20210101); **G02B13/00** (20060101); **G02B13/02** (20060101); **G02B27/01** (20060101); **H04N23/55** (20230101); **H04N23/58** (20230101)

**U.S. Cl.:**

**CPC** **G02B7/021** (20130101); **G02B7/08** (20130101); **G02B13/004** (20130101);  
**G02B13/0065** (20130101); **G02B13/009** (20130101); **G02B13/02** (20130101);  
**G02B27/0149** (20130101); **H04N23/55** (20230101); **H04N23/58** (20230101);

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## Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of U.S. patent application Ser. No. 18/314,866 filed May 10, 2023 (now allowed), which was a continuation application of U.S. patent application Ser. No. 17/665,495 filed Feb. 5, 2022 (now U.S. Pat. No. 11,668,894), which was a continuation application of U.S. patent application Ser. No. 16/851,475 filed Apr. 17, 2020 (now U.S. Pat. No. 11,347,020), which was a continuation application of U.S. patent application Ser. No. 16/392,754 filed Apr. 24, 2019 (now U.S. Pat. No. 10,670,827), which was a continuation application of U.S. patent application Ser. No. 16/252,608 filed Jan. 19, 2019 (now U.S. Pat. No. 10,571,644), which was a continuation application of U.S. patent application Ser. No. 16/077,477 filed Aug. 12, 2018 (now U.S. Pat. No. 11,347,016), which was a 371 application from international patent application No. PCT/IB2018/050988 filed Feb. 18, 2018 and claims the benefit of U.S. Provisional patent applications No. 62/462,438 filed Feb. 23, 2017, 62/478,783 filed Mar. 30, 2017, and 62/483,422 filed Apr. 9, 2017, all of which are incorporated herein by reference in their entirety.

### TECHNICAL FIELD

[0002] The presently disclosed subject matter is generally related to the field of digital cameras.

### BACKGROUND

[0003] Dual-aperture zoom cameras (also referred to as dual-cameras), in which one camera (also referred to as “sub-camera”) has a Wide FOV (“Wide sub-camera”) and the other has a narrow FOV (“Tele sub-camera”), are known.

[0004] International patent publication WO 2016/024192, which is incorporated herein by reference in its entirety, discloses a “folded camera module” (also referred to simply as “folded camera”) that reduces the height of a compact camera. In the folded camera, an optical path folding element (referred to hereinafter as “OPFE”) e.g. a prism or a mirror (otherwise referred to herein collectively as “reflecting element”) is added in order to tilt the light propagation direction from perpendicular to the smart-phone back surface to parallel to the smart-phone back surface. If the folded camera is part of a dual-aperture camera, this provides a folded optical path through one lens assembly (e.g. a Tele lens). Such a camera is referred to herein as “folded-lens dual-aperture camera”. In general, the folded camera may be included in a multi-aperture camera, for example together with two “non-folded” (upright) camera modules in a triple-aperture camera.

## SUMMARY

[0005] A small height of a folded camera is important to allow a host device (e.g. a smartphone, tablets, laptops or smart TV) that includes it to be as thin as possible. The height of the camera is often limited by the industrial design. In contrast, increasing the optical aperture of the lens results in an increase in the amount of light arriving at the sensor and improves the optical properties of the camera.

[0006] Therefore, there is a need for, and it would be advantageous to have a folded camera in which the height of the lens optical aperture is maximal for a given camera height and/or for a lens module height.

[0007] According to some aspects of the presently disclosed subject matter, there are provided digital cameras comprising an optical lens module including  $N \geq 3$  lens elements  $L_{\text{sub}.i}$  having a first optical axis, each lens element comprising a respective front surface  $S_{\text{sub}.2i-1}$  and a respective rear surface  $S_{\text{sub}.2i}$ , the lens element surfaces marked  $S_{\text{sub}.k}$  where  $1 \leq k \leq 2N$ , an image sensor, and a reflecting element, inclined with respect to the first optical axis so as to provide a folded optical path between an object and the lens elements, wherein each lens element surface has a clear height value  $CH(S_{\text{sub}.k})$  and wherein a clear height value  $CH(S_{\text{sub}.1})$  of surface  $S_{\text{sub}.1}$  is greater than a clear height value of each of surfaces  $S_{\text{sub}.2}$  to  $S_{\text{sub}.2N}$ .

[0008] In an exemplary embodiment, the  $N$  lens elements have an axial symmetry.

[0009] In an exemplary embodiment,  $CH(S_{\text{sub}.1}) \geq 1.1 \times CH(S_{\text{sub}.2})$ .

[0010] In an exemplary embodiment,  $CH(S_{\text{sub}.1}) \geq 1.2 \times CH(S_{\text{sub}.k})$  for  $3 \leq k \leq 2N$ .

[0011] In an exemplary embodiment, the digital camera has a total track length TTL and a focal back length BFL and wherein  $BFL \geq 0.3 \times TTL$ .

[0012] In an exemplary embodiment,  $L_{\text{sub}.1}$  is made of glass.

[0013] In an exemplary embodiment,  $L_{\text{sub}.1}$  is made of plastic.

[0014] In an exemplary embodiment,  $L_{\text{sub}.i}$  is made of plastic for any  $2 \leq i \leq N$ .

[0015] In an exemplary embodiment, the optical lens module is a front aperture lens module.

[0016] In an exemplary embodiment,  $CH(S_{\text{sub}.1}) < 7$  mm.

[0017] In some exemplary embodiments, each respective lens element surface  $S_{\text{sub}.k}$  has a clear aperture value  $CA(S_{\text{sub}.k})$ . In an exemplary embodiment, clear aperture value  $CA(S_{\text{sub}.1})$  of surface  $S_{\text{sub}.1}$  is greater than a clear aperture value of each of surfaces  $S_{\text{sub}.2}$  to  $S_{\text{sub}.2N}$ . In an exemplary embodiment,  $CA(S_{\text{sub}.1})$  is equal to clear aperture value  $CA(S_{\text{sub}.2N})$ , and  $CA(S_{\text{sub}.1})$  is greater than  $CA(S_{\text{sub}.k})$  for  $2 \leq k \leq 2N-1$ .

[0018] In an exemplary embodiment,  $CA(S_{\text{sub}.1})$  is substantially equal to  $CH(S_{\text{sub}.1})$ .

[0019] In an exemplary embodiment,  $CA(S_{\text{sub}.1}) \geq 1.1 \times CA(S_{\text{sub}.2})$ .

[0020] In an exemplary embodiment,  $CA(S_{\text{sub}.1}) \geq 1.2 \times CH(S_{\text{sub}.k})$  for  $3 \leq k \leq 2N$ . [0021] 1. In an exemplary embodiment, at least some of the lens elements have a width  $W_{\text{sub}.L}$  greater than their height  $H_{\text{sub}.L}$ .

[0022] In some exemplary embodiments, the optical lens module includes a cavity that holds the plurality of lens elements, wherein the cavity comprises a first portion in which the first lens element  $L_{\text{sub}.1}$  is located and a second portion in which a least one of the other lens elements are located, wherein the height of the first portion is greater than the height of the second portion.

[0023] In some exemplary embodiments, the optical lens module includes a cavity that holds at least some of lens elements  $L_{\text{sub}.2}$  to  $L_{\text{sub}.N}$ , wherein first lens element  $L_{\text{sub}.1}$  is located outside of the optical lens module.

[0024] In some exemplary embodiments, the image sensor is a rectangular sensor or a circular sensor.

[0025] In some exemplary embodiments,  $N \leq 6$ .

[0026] According to an aspect of the presently disclosed subject matter, there is provided a digital dual-camera comprising a camera of any of the embodiment mentioned above, wherein the camera is a Tele sub-camera configured to provide a Tele image, and a Wide sub-camera configured to

provide a Wide image.

[0027] According to some aspects of the presently disclosed subject matter, there is provided a digital camera comprising an optical lens module including  $N \geq 3$  lens elements  $L_{\text{sub}.i}$  having a first optical axis, each lens element comprising a respective front surface  $S_{\text{sub}.2i-1}$  and a respective rear surface  $S_{\text{sub}.2i}$ , the lens element surfaces marked  $S_{\text{sub}.k}$  where  $1 \leq k \leq 2N$ , an image sensor, and a reflecting element inclined with respect to the first optical axis so as to provide a folded optical path between an object and the lens elements, wherein each lens element surface has a clear aperture value  $CA(S_{\text{sub}.k})$  and wherein clear aperture value  $CA(S_{\text{sub}.1})$  is greater than  $CA(S_{\text{sub}.k})$  for  $2 \leq k \leq 2N$ .

[0028] In an exemplary embodiment,  $CA(S_{\text{sub}.1}) \geq 1.1 \times CA(S_{\text{sub}.2})$ .

[0029] In an exemplary embodiment,  $CA(S_{\text{sub}.1}) \geq 1.2 \times CH(S_{\text{sub}.k})$ , for  $3 \leq k \leq 2N$ .

[0030] In some exemplary embodiments, the optical lens module includes a cavity that holds the plurality of lens elements, wherein a height of cavity, measured along an axis orthogonal to the first optical axis, is variable along the first optical axis.

[0031] In some exemplary embodiments, the cavity comprises a first portion in which the first lens element  $L_{\text{sub}.1}$  is located and a second portion at which at least one of the other lens elements is located, and wherein the height of the first portion of the cavity is greater than the height of the second portion of the cavity.

[0032] In some exemplary embodiments, the optical lens module further comprises a lens barrel (or simply “barrel”) with a cavity in which at least some of lens elements  $L_{\text{sub}.2}$  to  $L_{\text{sub}.N}$  are held and wherein lens element  $L_{\text{sub}.1}$  is located outside of the barrel.

[0033] According to another aspect of the presently disclosed subject matter, a camera described above is a Tele sub-camera configured to provide a Tele image and is included together with a Wide sub-camera configured to provide a Wide image in a dual-camera.

[0034] According to another aspect of the presently disclosed subject matter, there is provided a digital camera comprising  $N$  lens elements having a symmetry along a first optical axis, wherein  $N$  is equal to or greater than 3, an image sensor, a reflecting element operative to provide a folded optical path between an object and the image sensor, and a barrel with a cavity in which the plurality of lens elements are held, wherein a height of cavity, measured along an axis orthogonal to the first optical axis, is variable along the first optical axis, wherein the cavity comprises a first portion in which the first lens element  $L_{\text{sub}.1}$  is located and a second portion at which at least one of the other lens elements is located, and wherein the height of the first portion of the cavity  $H_{\text{sub}.1}$  is greater than the height of the second portion of the cavity  $H_{\text{sub}.2}$ , so that  $H_{\text{sub}.1} > 1.1 \times H_{\text{sub}.2}$ .

[0035] According to another aspect of the presently disclosed subject matter, there is provided a digital camera comprising  $N$  lens elements  $L_{\text{sub}.1}$  to  $L_{\text{sub}.N}$  having axial symmetry along a first optical axis, wherein  $N$  is equal to or greater than 3, an image sensor, a reflecting element operative to provide a folded optical path between an object and the image sensor, and a barrel with a cavity in which at least some of the lens elements  $L_{\text{sub}.2}$  to  $L_{\text{sub}.N}$  are held, and wherein the lens element  $L_{\text{sub}.1}$  is located outside of barrel.

[0036] In an exemplary embodiment,  $L_{\text{sub}.N}$  is located outside the barrel.

[0037] According to some aspects of the presently disclosed subject matter, there is provided an optical lens module comprising a barrel having a cavity surrounded by walls and  $N$  lens elements  $L_{\text{sub}.1}$  to  $L_{\text{sub}.N}$ , wherein  $N$  is equal to or greater than 3, wherein  $L_{\text{sub}.1}$  has a portion which is not completely surrounded by the cavity and wherein walls of the cavity are aligning a center of lens element  $L_{\text{sub}.1}$  with the first optical axis.

[0038] In an exemplary embodiment,  $L_{\text{sub}.N}$  has a portion which is not completely surrounded by the cavity and wherein walls of the cavity are aligning a center of lens element  $L_{\text{sub}.N}$  with the first optical axis.

[0039] In an exemplary embodiment, at least one of an extremity of the walls and an extremity of

lens element L.sub.1 is shaped so that the extremity of the walls acts a stop for at least a portion of lens element L.sub.1, thereby substantially aligning a center of lens element L.sub.1 with the first optical axis.

[0040] In an exemplary embodiment, a first portion of lens element L.sub.1 is located in the cavity between the extremity of the walls and a second portion of lens element L.sub.1 is located outside the barrel and wherein a thickness of the second portion of lens element L.sub.1 along the first optical axis is greater than a thickness of the first portion of lens element L.sub.1 along the first optical axis.

[0041] In an exemplary embodiment, a cross-section of the extremity of the walls has a stepped shape.

[0042] In an exemplary embodiment, a cross-section of the extremity of lens element L.sub.1 has a stepped shape.

[0043] In an exemplary embodiment, a cross-section of the extremity of the walls has a sloping shape.

[0044] In an exemplary embodiment, the extremity of the walls comprises a chamfer.

[0045] In an exemplary embodiment, the lens module further comprises a cover for protecting the lens, the cover covering lens element L.sub.1.

[0046] In an exemplary embodiment, the cover has an extreme point beyond lens element L.sub.1.

[0047] In an exemplary embodiment, the cover blocks light from entering a mechanical part of lens element L.sub.1.

[0048] According to some aspects of the presently disclosed subject matter, there is provided an optical lens module comprising a plurality  $N \geq 3$  of lens elements L.sub.i wherein  $1 \leq i \leq N$ , each lens element comprising a respective front surface S.sub.2i-1 and a respective rear surface S.sub.2i, the lens element surfaces marked S.sub.k where  $1 \leq k \leq 2N$ , wherein each lens element surface has a clear aperture value CA(S.sub.k), wherein CA(S.sub.1) is substantially equal to CA(S.sub.2N) and wherein CA(S.sub.1) is greater CA(S.sub.k) for  $2 \leq k \leq 2N-1$ .

[0049] According to some aspects of the presently disclosed subject matter, there is provided an optical lens module comprising a plurality  $N \geq 3$  of lens elements L.sub.i wherein  $1 \leq i \leq N$ , each lens element comprising a respective front surface S.sub.2i-1 and a respective rear surface S.sub.2i, the lens element surfaces marked S.sub.k where  $1 \leq k \leq 2N$ , wherein each lens element surface has a clear aperture value CA(S.sub.k) and wherein CA(S.sub.1) is greater CA(S.sub.k) for  $2 \leq k \leq 2N$ .

[0050] According to some aspects of the presently disclosed subject matter, there is provided a digital camera comprising an image sensor, a reflecting element operative to provide a folded optical path between an object and the image sensor, and an optical lens module as described above.

[0051] According to another aspect of the presently disclosed subject matter, there is provided an optical lens module comprising a barrel having a barrel height H and N lens elements L.sub.1 to L.sub.N, wherein N is equal to or greater than 3 and wherein a height H.sub.L1 of lens element L.sub.1 satisfies or fulfills  $H_{\text{sub.L1}} \geq H$ . In an exemplary embodiment,  $H_{\text{sub.LN}} \geq H$ . In an exemplary embodiment,  $H_{\text{sub.LN}} = H_{\text{sub.L1}}$ .

[0052] According to another aspect of the presently disclosed subject matter, there is provided an optical lens module comprising N lens elements L.sub.1 to L.sub.N, wherein  $N \geq 3$ , wherein each lens element L.sub.i has a height H.sub.Li for  $1 \leq i \leq N$  and wherein  $H_{\text{sub.L1}} \geq H_{\text{sub.LN}} > H_{\text{sub.L2}}$ .

[0053] In an exemplary embodiment,  $H_{\text{sub.L1}} > H_{\text{sub.Li}}$  for  $3 \leq i \leq N-1$ .

[0054] According to another aspect of the presently disclosed subject matter, there is provided a method for assembling an optical lens module, comprising: inserting a first lens element L.sub.1 of N lens elements into a barrel from an object side of the barrel, fixedly attaching lens element L.sub.1 to the barrel, inserting other lens elements L.sub.2 to L.sub.N and spacers R.sub.1 to R.sub.N that separate respective lens elements from an image side of the barrel in an insertion order R.sub.1, L.sub.2 . . . R.sub.N-1, L.sub.N, and fixedly attaching lens element L.sub.N to the lens

module.

[0055] According to another aspect of the presently disclosed subject matter, there is provided a mobile electronic device comprising an internal digital camera integrated inside a housing of the mobile electronic device, wherein the digital camera is in accordance with any one of the aspects mentioned above, or comprises any of the optical lens module described above.

[0056] According to another aspect of the presently disclosed subject matter there is provided a multiple-aperture camera comprising at least one Wide sub-camera and at least one Tele sub-camera, which is in accordance with any one of the aspects mentioned above, or comprises any of the optical lens module described above.

[0057] According to another aspect of the presently disclosed subject matter, the reflecting element is a rotating reflecting element capable of being rotated around one or two axes in order to move the position of a field of view (FOV) of the digital camera and capture a plurality of adjacent non-overlapping or partially overlapping images at a plurality of respective positions, and the digital camera is configured to generate from the plurality of images a composite image having an overall image FOV greater than an FOV of the digital camera.

[0058] In some exemplary embodiment, the digital camera according to the above aspect further comprises an actuator configured to apply rotational movement around one or two axes to the rotating reflecting element, the actuator operatively connected to a controller configured to control the actuator to cause the camera to scan an area corresponding to a requested zoom factor, the area being greater than the FOV of the digital camera, and to capture the plurality of images where each image is captured at a different position in the scanned area.

[0059] In an exemplary embodiment, the size of the composite image is generated by the stitching of 4 Tele images.

[0060] In an exemplary embodiment, the size of the composite image is generated by the stitching of 6 Tele images.

[0061] In an exemplary embodiment, the size of the composite image is generated by the stitching of 9 Tele images.

[0062] In an exemplary embodiment, a combined size of the plurality of image is greater than the size of the composite image.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0063] Non-limiting examples of embodiments disclosed herein are described below with reference to figures attached hereto that are listed following this paragraph. The drawings and descriptions are meant to illuminate and clarify embodiments disclosed herein, and should not be considered limiting in any way. Like elements in different drawings may be indicated by like numerals.

Elements in the drawings are not necessarily drawn to scale. In the drawings:

[0064] FIG. 1A is a general isometric view of an example of a known folded camera;

[0065] FIG. 1B is a side view of the camera of FIG. 1A;

[0066] FIG. 1C is a general isometric view of an example of a known camera comprising a folded Tele sub-camera and a Wide sub-camera;

[0067] FIG. 1D is a side view of the camera of FIG. 1C;

[0068] FIG. 2A is a schematic view of one embodiment of lens elements with light rays according to some examples of the presently disclosed subject matter;

[0069] FIG. 2B is another schematic view of the lens elements of FIG. 2A;

[0070] FIG. 3A is a schematic view of impact points of optical rays impinging a convex surface of a lens element, and a schematic view of the orthogonal projection of the impact points on a plane P, according to some examples of the presently disclosed subject matter;

[0071] FIG. 3B is a schematic view of impact points of optical rays impinging a concave surface of a lens element, and a schematic view of the orthogonal projection of the impact points on a plane P, according to some examples of the presently disclosed subject matter;

[0072] FIG. 4 is a schematic representation of the orthogonal projection of the impact points on a plane P, and of a clear height value (“CH”), according to some examples of the presently disclosed subject matter;

[0073] FIG. 5 is a schematic representation of the orthogonal projection of the impact points on a plane P, and of a clear aperture value (“CA”), according to some examples of the presently disclosed subject matter;

[0074] FIG. 6 is a schematic representation of a side view of an optical lens module for holding the lens elements, according to some examples of the presently disclosed subject matter;

[0075] FIG. 7 is a schematic representation of a side view of an optical lens module for holding the lens elements, according to other examples of the presently disclosed subject matter;

[0076] FIG. 8 is a schematic representation of an example of an optical lens module comprising a plurality of lens elements, according to the presently disclosed subject matter;

[0077] FIG. 9A is a schematic representation of another example of an optical lens module comprising a plurality of lens elements, according to the presently disclosed subject matter;

[0078] FIG. 9B depicts a variant of the example of FIG. 9A;

[0079] FIG. 10 is a schematic representation of another example of an optical lens module comprising a plurality of lens elements, according to the presently disclosed subject matter;

[0080] FIG. 11A is a schematic representation of an isometric view of a barrel and of a plurality of lens elements before their insertion into the barrel, according to the presently disclosed subject matter;

[0081] FIG. 11B depicts a cross-section view of the example of FIG. 11A, along plane Y-Z;

[0082] FIG. 11C depicts a cross-section view of the example of FIG. 11A, along plane X-Z;

[0083] FIG. 11D depicts a front view of the example of FIG. 11A;

[0084] FIG. 11E depicts another isometric view of the example of FIG. 11A after the insertion of the lens elements in the barrel;

[0085] FIG. 11F is a schematic representation of a front view of a lens element;

[0086] FIG. 12 is a schematic representation of a manufacturing process of the optical lens module of FIG. 11A to FIG. 11E.

[0087] FIG. 13A is a schematic representation of an isometric view of a plurality of lens elements;

[0088] FIG. 13B is a schematic representation of an isometric view of an optical lens module comprising the plurality of lens elements of FIG. 13A and a barrel;

[0089] FIG. 13C is yet another schematic representation of an isometric view of an optical lens module comprising the plurality of lens elements of FIG. 13A and a barrel.

[0090] FIG. 14 is a schematic illustration of a stitched image generated by capturing and stitching together 4 Tele images, according to some examples of the presently disclosed subject matter;

[0091] FIG. 15 is a schematic illustration of a stitched image generated by capturing and stitching together 6 Tele images, according to some examples of the presently disclosed subject matter;

[0092] FIG. 16 is a schematic illustration of a stitched image generated by capturing and stitching together 9 Tele images, according to some examples of the presently disclosed subject matter;

[0093] FIG. 17A shows an isometric view of another embodiment of a barrel with lens elements, according to the presently disclosed subject matter;

[0094] FIG. 17B shows a side cut of the barrel and lens elements of FIG. 17A;

[0095] FIG. 17C shows an exploded view of the lens elements of FIG. 17B;

[0096] FIG. 17D shows a side cut of another barrel with lens elements, according to the presently disclosed subject matter;

[0097] FIG. 18A shows an isometric view of yet another embodiment of a lens module with a barrel and lens elements, according to the presently disclosed subject matter;

[0098] FIG. **18B** shows a side cut of the lens module of FIG. **18A**;  
[0099] FIG. **18C** shows an exploded view of the lens module of FIG. **18B**;  
[0100] FIG. **19A** shows a side cut of yet another embodiment of a lens module with a barrel and lens elements, according to the presently disclosed subject matter;  
[0101] FIG. **19B** shows the lens module of FIG. **19A** in an exploded view;  
[0102] FIG. **20** shows a side cut of yet another embodiment of a barrel with lens elements, according to the presently disclosed subject matter;  
[0103] FIG. **21A** is a schematic view of another embodiment of lens elements showing light rays, according to another example of the presently disclosed subject matter;  
[0104] FIG. **21B** is another schematic view of the lens elements of FIG. **21A**;  
[0105] FIG. **22** is a schematic representation of a side view of optical lens module for holding the lens elements of FIGS. **21A**, **21B**;  
[0106] FIG. **23** is a schematic representation of a side view of another optical lens module for holding the lens elements of FIGS. **21A**, **21B**;  
[0107] FIG. **24** is a schematic representation of yet another example of an optical lens module comprising a plurality of lens elements, according to the presently disclosed subject matter;  
[0108] FIG. **25A** is a schematic representation of an isometric view of another optical lens module, according to the presently disclosed subject matter;  
[0109] FIG. **25B** depicts a cross-section view of the lens module of FIG. **25A**, along plane Y-Z;  
[0110] FIG. **25C** depicts a cross-section view of the lens module of FIG. **25A**, along plane X-Z;  
[0111] FIG. **25D** depicts another isometric view of the lens module of FIG. **25A** after the insertion of the lens elements into the barrel;  
[0112] FIG. **26A** shows an isometric view of yet another embodiment of a lens module with a barrel and lens elements, according to the presently disclosed subject matter;  
[0113] FIG. **26B** shows a side cut of lens module of FIG. **26A**;  
[0114] FIG. **26C** shows an exploded view of the lens module of FIG. **26B**;  
[0115] FIG. **27** shows an isometric view of yet another embodiment of a lens module, according to the presently disclosed subject matter;

#### DETAILED DESCRIPTION

[0116] In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding. However, it will be understood by those skilled in the art that the presently disclosed subject matter may be practiced without these specific details. In other instances, well-known methods have not been described in detail so as not to obscure the presently disclosed subject matter.

[0117] It is appreciated that certain features of the presently disclosed subject matter, which are, for clarity, described in the context of separate embodiments, may also be provided in combination in a single embodiment. Conversely, various features of the presently disclosed subject matter, which are, for brevity, described in the context of a single embodiment, may also be provided separately or in any suitable sub-combination.

[0118] The term “processing unit” as disclosed herein should be broadly construed to include any kind of electronic device with data processing circuitry, which includes for example a computer processing device operatively connected to a computer memory (e.g. digital signal processor (DSP), a microcontroller, a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), etc.) capable of executing various data processing operations.

[0119] Furthermore, for the sake of clarity the term “substantially” is used herein to imply the possibility of variations in values within an acceptable range. According to one example, the term “substantially” used herein should be interpreted to imply possible variation of up to 10% over or under any specified value. According to another example, the term “substantially” used herein should be interpreted to imply possible variation of up to 5% over or under any specified value. According to a further example, the term “substantially” used herein should be interpreted to imply



possible variation of up to 2.5% over or under any specified value.

[0120] FIGS. 1A and 1B illustrate a known digital folded camera **100**, which may operate for example as a Tele camera. Digital camera **100** comprises a first reflecting element (e.g. mirror or prism, and also referred to sometimes as “optical path folding element” (OPFE)) **101**, a plurality of lens elements (not visible in this representation, but visible e.g. in FIGS. 2A and 2B) and an image sensor **104**. The lens elements (and also barrel, the optical lens module) may have axial symmetric along a first optical axis **103**. At least some of the lens elements can be held by a structure called a “barrel” **102**. An optical lens module comprises the lens elements and the barrel. The barrel can have a longitudinal symmetry along optical axis **103**. In FIGS. 1A to 1D, the cross-section of this barrel is circular. This is however not mandatory and other shapes can be used.

[0121] The path of the optical rays from an object (not shown) to image sensor **104** defines an optical path (see optical paths **105** and **106**, which represent portions of the optical path).

[0122] OPFE **101** may be a prism or a mirror. As shown in FIG. 1A, OPFE **101** can be a mirror inclined with respect to optical axis **103**. In other cases (not shown, see for example PCT/IB2017/052383), OPFE **101** can be a prism with a back surface inclined with respect to optical axis **103**. OPFE folds the optical path from a first optical path **105** to a second optical path **106**. Optical path **106** is substantially parallel to the optical axis **103**. The optical path is thus referred to as “folded optical path” (indicated by optical paths **105** and **106**) and camera **100** is referred to as “folded camera”. The lens module comprises the plurality of lens elements.

[0123] In particular, in some examples, OPFE **101** can be inclined at substantially 45° with respect to optical axis **103**. In FIG. 1A, OPFE **101** is also inclined at substantially 45° with respect to optical path **105**.

[0124] In some known examples, image sensor **104** lies in a X-Y plane substantially perpendicular to optical axis **103**. This is however not limiting and the image sensor **104** can have a different orientation. For example, and as described in WO 2016/024192, image sensor **104** can be in the XZ plane. In this case, an additional OPFE can be used to reflect the optical rays towards image sensor **104**.

[0125] According to some examples, image sensor **104** has a rectangular shape. According to some examples, image sensor **104** has a circular shape. These examples are however not limiting.

[0126] In various examples camera **100** may be mounted on a substrate **109**, e.g. a printed circuit board (PCB), as known in the art.

[0127] Two sub-cameras, for example a regular Wide sub-camera **130** and a Tele sub-camera **100** may be included in a digital camera **170** (also referred to as dual-camera or dual-aperture camera). A possible configuration is described with reference to FIGS. 1C and 1D. In this example, Tele sub-camera **100** is according to the camera described with reference to FIGS. 1A and 1B. The components of Tele sub-camera **100** thus have the same reference numbers as in FIGS. 1A and 1B, and are not described again.

[0128] Wide camera **130** can include an aperture **132** (indicating object side of the camera) and an optical lens module **133** (or “Wide lens module” in short) with a symmetry (and optical) axis **134** in the Y direction, as well as a Wide image sensor **135**. The Wide sub-camera comprises a Wide lens module configured to provide a Wide image, wherein the Wide sub-camera has a Wide field of view (FOV.sub.W) and the Tele sub-camera has a Tele field of view (FOV.sub.T) narrower than FOV.sub.W. Notably, in other examples a plurality of Wide sub-cameras and/or a plurality of Tele sub-cameras can be incorporated and operative in a single digital camera.

[0129] According to one example, the Wide image sensor **135** lies in the X-Z plane, while image sensor **104** (which is in this example is a Tele image sensor) lies in a X-Y plane substantially perpendicular to optical axis **103**.

[0130] In the examples of FIGS. 1A to 1D, camera **100** can further include (or be otherwise operatively connected to) a processing unit comprising one or more suitably configured processors (not shown) for performing various processing operations, for example processing the Tele image

and the Wide image into a fused output image.

[0131] The processing unit may include hardware (HW) and software (SW) specifically dedicated for operating with the digital camera. Alternatively, a processor of an electronic device (e.g. its native CPU) in which the camera is installed can be adapted for executing various processing operations related to the digital camera (including, but not limited to, processing the Tele image and the Wide image into an output image).

[0132] Attention is now drawn to FIGS. 2A and 2B, which show schematic view of a lens module **200** having lens elements shown with optical rays according to some examples of the presently disclosed subject matter. Lens module **200** is shown without a lens barrel. FIG. 2A shows optical ray tracing of lens module **200** while FIG. 2B shows only the lens elements for more clarity. In addition, both figures show an image sensor **202** and an optical element **205**.

[0133] Lens module **200** includes a plurality of  $N$  lens elements  $L_{\text{sub}.i}$  (wherein “ $i$ ” is an integer between 1 and  $N$ ).  $L_{\text{sub}.1}$  is the lens element closest to the object side and  $L_{\text{sub}.N}$  is the lens element closest to the image side, i.e. the side where the image sensor is located. This order holds for all lenses and lens elements disclosed herein. Lens elements  $L_{\text{sub}.i}$  can be used e.g. as lens elements of camera **100** represented in FIGS. 1A and 1B or as lens elements of the Tele sub-camera **100** of FIGS. 1C and 1D. As shown, the  $N$  lens elements are axial symmetric along optical axis **103**.

[0134] In the examples of FIGS. 2A and 2B,  $N$  is equal to four. This is however not limiting and a different number of lens elements can be used. According to some examples,  $N$  is equal to or greater than 3. For example,  $N$  can be equal to 3, 4, 5, 6 or 7.

[0135] In the examples of FIGS. 2A and 2B, some of the surfaces of the lens elements are represented as convex, and some are represented as concave. The representation of FIGS. 2A and 2B is however not limiting and a different combination of convex and/or concave surfaces can be used, depending on various factors such as the application, the desired optical power, etc.

[0136] Optical rays (after their reflection by a reflecting element, such as OPFE **101**) pass through lens elements  $L_{\text{sub}.i}$  and form an image on an image sensor **202**. In the examples of FIGS. 2A and 2B, the optical rays pass through an optical element **205** (which comprises a front surface **205a** and a rear surface **205b**, and can be e.g. a cut-off filter) before impinging on image sensor **202**. This is however not limiting, and in some examples, optical element **205** is not present. Optical element **205** may be for example infra-red (IR) filter, and/or a glass image sensor dust cover.

[0137] Each lens element  $L_{\text{sub}.i}$  comprises a respective front surface  $S_{\text{sub}.2i-1}$  (the index “ $2i-1$ ” being the number of the front surface) and a respective rear surface  $S_{\text{sub}.2i}$  (the index “ $2i$ ” being the number of the rear surface), where “ $i$ ” is an integer between 1 and  $N$ . This numbering convention is used throughout the description. Alternatively, as done throughout this description, lens surfaces are marked as “ $S_{\text{sub}.k}$ ”, with  $k$  running from 1 to  $2N$ . The front surface and the rear surface can be in some cases aspherical. This is however not limiting.

[0138] As used herein the term “front surface” of each lens element refers to the surface of a lens element located closer to the entrance of the camera (camera object side) and the term “rear surface” refers to the surface of a lens element located closer to the image sensor (camera image side).

[0139] As explained below, a clear height value  $CH(S_{\text{sub}.k})$  can be defined for each surface  $S_{\text{sub}.k}$  for  $1 \leq k \leq 2N$ , and a clear aperture value  $CA(S_{\text{sub}.k})$  can be defined for each surface  $S_{\text{sub}.k}$  for  $1 \leq k \leq 2N$ .  $CA(S_{\text{sub}.k})$  and  $CH(S_{\text{sub}.k})$  define optical properties of each surface  $S_{\text{sub}.k}$  of each lens element.

[0140] In addition, and as shown e.g. in FIG. 6, a height (“ $H_{\text{sub}.Li}$ ”, for  $1 \leq i \leq N$ ) is defined for each lens element  $L_{\text{sub}.i}$ .  $H_{\text{sub}.Li}$  corresponds, for each lens element  $L_{\text{sub}.i}$ , to the maximal height of lens element  $L_{\text{sub}.i}$  measured along an axis perpendicular to the optical axis of the lens elements (in the example in FIG. 6,  $H_{\text{sub}.Li}$  is measured along optical path **105** which is perpendicular to optical axis **103**). For a given lens element, the height is greater than, or equal to the clear height

value  $CH$  and the clear aperture value  $CA$  of the front and rear surfaces of this given lens element. Typically, for an axial symmetric lens element,  $H_{sub.Li}$  is the diameter of lens element  $L_{sub.i}$  as seen in FIG. 11F. Typically, for an axial symmetric lens element,  $H_{sub.Li} = \max\{CA(S_{sub.2i-1}), CA(S_{sub.2i})\} + \text{mechanical part size}$ . The mechanical part and its properties are defined below (FIGS. 11E, 11F and 17A-D). The mechanical part size contribution to  $H_{sub.Li}$  is typically 200  $\mu\text{m}$ -1000  $\mu\text{m}$ .

[0141] In addition, as also shown in FIG. 6, a height  $H$  is defined for the lens barrel. For any axis  $A$  which is perpendicular to the optical axis of a lens module, a diameter  $D_{sub.A}$  is defined as the maximal distance measured along axis  $A$  of the lens module.  $H$  is defined as the minimum of all  $D_{sub.As}$  for all possible axes  $A$ . In the example in FIG. 6,  $H$  corresponds to the maximal height of the barrel measured along an axis perpendicular to the optical axis **103** of the lens module and parallel to optical path **105**.

[0142] In addition, as also shown in FIG. 7, a height  $H_{sub.C}$  is defined for the cavity of a lens barrel.  $H_{sub.C}$  corresponds to the height of the cavity barrel measured along an axis perpendicular to the optical axis of the lens module (in the example in FIG. 7,  $H_{sub.C}$  is measured along optical path **105** which is perpendicular to optical axis **103**). In some examples, where the cavity barrel is axial-symmetric,  $H_{sub.C}$  is the internal diameter of the cavity barrel.

[0143] According to some examples of the presently disclosed subject matter, the closest lens element to the object side ( $L_{sub.1}$ ) has a height which is greater than the height of each of the other lens elements. A non-limiting example is shown in FIG. 6, where  $H_{sub.L1}$  is greater than  $H_{sub.L2}$ ,  $H_{sub.L3}$  and  $H_{sub.L4}$ .

[0144] According to some examples of the presently disclosed subject matter, the closest lens element to the object side ( $L_{sub.1}$ ) and the closest lens element to the image sensor ( $L_{sub.N}$ ) have a height which is substantially equal and is greater than the height of each of the other lens elements. A non-limiting example is shown in FIG. 21B, where  $H_{sub.L1}$  equal to  $H_{sub.L5}$  and are greater than  $H_{sub.L2}$ ,  $H_{sub.L3}$  and  $H_{sub.L4}$ .

[0145] As shown in FIGS. 3A, 3B and 4, each optical ray that passes through a surface  $S_{sub.k}$  (for  $1 \leq k \leq 2N$ ) impinges this surface on an impact point  $IP$ . Optical rays enter lens module **200** from surface  $S_{sub.1}$ , and pass through surfaces  $S_{sub.2}$  to  $S_{sub.2N}$  consecutively. Some optical rays can impinge on any surface  $S_{sub.k}$  but cannot/will not reach image sensor **202**. For a given surface  $S_{sub.k}$ , only optical rays that can form an image on image sensor **202** are considered forming a plurality of impact points  $IP$  are obtained.  $CH(S_{sub.k})$  is defined as the distance between two closest possible parallel lines (see lines **400** and **401** in FIG. 4 located on a plane  $P$  orthogonal to the optical axis of the lens elements (in the representation of FIGS. 3A and 3B, plane  $P$  is parallel to plane  $X$ - $Y$  and is orthogonal to optical axis **103**), such that the orthogonal projection  $IP_{sub.orth}$  of all impact points  $IP$  on plane  $P$  is located between the two parallel lines.  $CH(S_{sub.k})$  can be defined for each surface  $S_{sub.k}$  (front and rear surfaces, with  $1 \leq k \leq 2N$ ).

[0146] The definition of  $CH(S_{sub.k})$  does not depend on the object currently imaged, since it refers to the optical rays that “can” form an image on the image sensor. Thus, even if the currently imaged object is located in a black background which does not produce light, the definition does not refer to this black background since it refers to any optical rays that “can” reach the image sensor to form an image (for example optical rays emitted by a background which would emit light, contrary to a black background).

[0147] For example, FIG. 3A illustrates the orthogonal projections  $IP_{sub.orth,1}$ ,  $IP_{sub.orth,2}$  of two impact points  $IP_{sub.1}$  and  $IP_{sub.2}$  on plane  $P$  which is orthogonal to optical axis **103**. By way of example, in the representation of FIG. 3A, surface  $S_{sub.k}$  is convex.

[0148] FIG. 3B illustrates the orthogonal projections  $IP_{sub.orth,3}$ ,  $IP_{sub.orth,4}$  of two impact points  $IP_{sub.3}$  and  $IP_{sub.4}$  on plane  $P$ . By way of example, in the representation of FIG. 3B, surface  $S_{sub.k}$  is concave.

[0149] In FIG. 4, the orthogonal projection  $IP_{sub.orth}$  of all impact points  $IP$  of a surface  $S_{sub.k}$

on plane P is located between parallel lines **400** and **401**. CH(S.sub.k) is thus the distance between lines **400** and **401**.

[0150] Attention is drawn to FIG. 5. According to the presently disclosed subject matter, a clear aperture CA(S.sub.k) is defined for each given surface S.sub.k (for  $1 \leq k \leq 2N$ ), as the diameter of a circle, wherein the circle is the smallest possible circle located in a plane P orthogonal to the optical axis **103** and encircling all orthogonal projections IP.sub.orth of all impact points on plane P. As mentioned above with respect to CH(S.sub.k), it is noted that the definition of CA(S.sub.k) also does not depend on the object which is currently imaged.

[0151] As shown in FIG. 5, the circumscribed orthogonal projection IP.sub.orth of all impact points IP on plane P is circle **500**. The diameter of this circle **500** defines CA(S.sub.k).

[0152] Detailed optical data and surface data are given in Tables 1-2 for the example of the lens elements in FIG. 2A-2B, in Tables 3 and 4 for the example of the lens elements in FIGS. 6-9, in Tables 5 and 6 for the example of the lens elements in FIG. 20 and in Tables 7 and 8 for the example of the lens elements in FIG. 21A-21B (see below). The values provided for these examples are purely illustrative and according to other examples, other values can be used.

[0153] In the tables below, the units of the radius of curvature ("R"), the lens element thickness ("Thickness") and the clear aperture ("CA") are expressed in millimeters.

[0154] Line "0" of Tables 1, 3 and 5 and 7 describes parameters associated to the object (not visible in the figures); the object is being placed at 1 km from the system, considered to be an infinite distance.

[0155] Lines "1" to "8" of Tables 1 to 4 describe respectively parameters associated to surfaces S.sub.1 to S.sub.8. Lines "1" to "10" of Tables 5 to 8 describe respectively parameters associated with surfaces S.sub.1 to S.sub.10.

[0156] Lines "9", "10" and "11" of Tables 1 and 3, and lines "11", "12" and "13" in Tables 5 and 7 describe respectively parameters associated to surfaces **205a**, **205b** of optical element **205** and of a surface **202a** of the image sensor **202**.

[0157] In lines "i" of Tables 1, 3 and 5 (with i between 1 and 10 in tables 1 and 3 and i between 1 and 12 in Table 5), the thickness corresponds to the distance between surface S.sub.i and surface S.sub.i+1, measured along the optical axis **103** (which coincides with the Z axis).

[0158] In line "11" of Tables 1, 3 (line "13" in Tables 5 and 7), the thickness is equal to zero, since this corresponds to the last surface **202a**.

[0159] "BK7", "K26R", "EP6000" and "H-ZK3" are conventional materials which are known to a person skilled in the art and which are mentioned by way of example.

[0160] "BK7" is characterized by the approximate following parameters: [0161] Refractive index of 1.5168, and [0162] Abbe number of 64.16733.

[0163] "K26R" is a material manufactured by Zeon Corporation, and is characterized by the approximate following parameters: [0164] Refractive index of 1.534809, and [0165] Abbe number of 55.663857.

[0166] "EP6000" is a material manufactured by Mitsubishi, and is characterized by the approximate following parameters: [0167] Refractive index of 1.6397, and [0168] Abbe number of 23.5288.

[0169] "H-ZK3" is a type of glass characterized by the approximate following parameters: [0170] Refractive index of 1.5891, and [0171] Abbe number of 61.25.

In Table 7, the properties of each surface material are given, with "Nd" as refractive index and "Vd" as Abbe number.

[0172] The equation of the surface profiles of each surface S.sub.k (for k between 1 and 2N) is expressed by:

$$[00001]z = \frac{cr^2}{1 + \sqrt{1 - (1+k)c^2r^2}} + A_1 r^4 + A_2 r^6 + A_3 r^8 + A_4 r^{10} + A_5 r^{12} + A_6 r^{14} + A_7 r^{16}$$

where "z" is the position of the profile of the surface S.sub.k measured along optical axis **103**

(coinciding with the Z axis, wherein  $z=0$  corresponds to the intersection of the profile of the surface S.sub.k with the Z axis), “r” is the distance from optical axis **103** (measured along an axis which is perpendicular to optical axis **103**), “K” is the conic coefficient,  $c=1/R$  where R is the radius of curvature, and A.sub.n (n from 1 to 7) are coefficients given in Tables 2 and 4 for each surface S.sub.k. The maximum value of r, “max r”, is equal to D/2.

[0173] In the example of FIGS. 2A and 2B, the following optical properties are achieved: [0174] TTL=13.6 mm. [0175] BFL=4.93 mm. [0176] EFL (effective focal length)=13.8 mm. [0177] CA(S.sub.1)=CH(S.sub.1)=5 mm. [0178] CA(S.sub.2)=CH(S.sub.2)=4.4 mm. [0179] For k between 3 and 8, CA(S.sub.k)≤3.8 mm, CH(S.sub.k)≤CA(S.sub.k). [0180] f/#=2.76 [0181] Focal length of L.sub.1: f.sub.1=5.57 mm, f.sub.1/EFL=0.4 [0182] Sensor diagonal (SD) 5.86 mm, last surface clear aperture CA(S.sub.2N)=3.8 mm, CA(S.sub.2N)/SD=0.65.

[0183] In the example of FIG. 6, the following optical properties are achieved: [0184] TTL=11.1 mm, [0185] BFL=4.3 mm, [0186] EFL (effective focal length)=11.2 mm, [0187] CA(S.sub.1)=CH(S.sub.1)=4.32 mm, [0188] CA(S.sub.2)=CH(S.sub.2)=3.52 mm, and [0189] For k between 3 and 8, CA(S.sub.k)≤3.2 mm, CH(S.sub.k)≤CA(S.sub.k). [0190] f/#=2.5 [0191] Focal length of L.sub.1: f.sub.1=4.54 mm, f.sub.1/EFL=0.4 [0192] Sensor diagonal (SD) 5.24 mm, last surface clear aperture CA(S.sub.2N)=3.2 mm, CA(S.sub.2N)/SD=0.61.

[0193] In the example of FIG. 20, the following optical properties are achieved: [0194] TTL=15 mm. [0195] BFL=6.9 mm. [0196] EFL=16 mm. [0197] CA(S.sub.1)=CH(S.sub.1)=5.92 mm. [0198] CA(S.sub.2)=CH(S.sub.2)=5.1 mm. [0199] For k between 3 and 10, CA(S.sub.k)≤4.0 mm, CH(S.sub.k)≤CA(S.sub.k). [0200] f/#=2.7 [0201] Focal length of L.sub.1: f.sub.1=8.1 mm, f.sub.1/EFL=0.506 [0202] Sensor diagonal (SD) 5.86 mm, last surface clear aperture CA(S.sub.2N)=3.52 mm, CA(S.sub.2N)/SD=0.6

[0203] In the example of FIGS. 21A and 21B, the following optical properties are achieved: [0204] TTL=7.78 mm. [0205] BFL=3.23 mm. [0206] EFL (effective focal length)=7.97 mm. [0207] CA(S.sub.1)=CH(S.sub.1)=3.6 mm. [0208] CA(S.sub.2)=CH(S.sub.2)=3.45 mm. [0209] For k between 3 and 8, CA(S.sub.k)≤3.4 mm, CH(S.sub.k)≤CA(S.sub.k). [0210] CA(S.sub.2N-1)=3.36 mm, CH(S.sub.2N-1)=2.842 mm [0211] CA(S.sub.2N)=3.6 mm, CH(S.sub.2N-1)=3.064 mm [0212] f/#=2.2 [0213] Focal length of L.sub.1: f.sub.1=3.972 mm, f.sub.1/EFL=0.498 [0214] Sensor diagonal (SD) 5.86 mm, CA(S.sub.2N)/SD=0.615.

In this application and for the properties above, the following symbols and abbreviations are used, all of which are terms known in the art: [0215] TTL: The “total track length” is defined as the maximal distance, measured along an axis parallel to the optical axis, between a point of the front surface S.sub.1 of the first lens element L.sub.1 and the image sensor, when the system is focused to an infinity object distance. [0216] BFL: The “focal back length” is defined as the minimal distance, measured along an axis parallel to the first optical axis, between a point of the rear surface S.sub.2N of the last lens element L.sub.N and the image sensor, when the system is focused to an infinity object distance. [0217] EFL: Effective focal length of a lens module (assembly of lens elements L.sub.1 to L.sub.N) [0218] f/#: f-number, the ratio of the EFL to the aperture stop diameter. [0219] Aperture stop: the opening which limits the amount of light which passes through an optical system.

TABLE-US-00001 TABLE 1 # R Thickness Material CA/2 Conic coefficient K 0 Infinity  
1.00E+06 1 4.018 3.122 K26R 2.50 -0.918 2 -8.544 0.427 2.20 -13.319 3 -11.602 0.383 EP6000  
1.90 68.256 4 4.252 0.668 1.90 0.035 5 12.410 3.072 EP6000 1.90 9.316 6 -9.884 0.565 1.90  
-50.842 7 -5.080 0.434 K26R 1.90 -30.682 8 -57.279 4.429 1.90 -207.271 9 Infinity 0.210 BK7  
10 Infinity 0.289 11 Infinity 0.000

TABLE-US-00002 TABLE 2 # A1 A2 A3 A4 A5 A6 A7 1 1.0982E-03 -5.6900E-05  
3.0019E-06 -3.0442E-07 -2.0532E-07 2.1748E-08 -2.5134E-09 2 1.4662E-03  
-6.8269E-04 3.6775E-05 1.2874E-07 -1.5311E-06 1.6528E-07 0.0000E+00 3  
-4.4641E-03 2.3303E-03 -6.0231E-04 5.0714E-05 2.4477E-06 -3.4785E-07 -1.2814E-08

4 -4.6819E-03 2.7039E-03 -4.9103E-04 -6.1960E-05 4.4187E-05 -5.1739E-06  
 0.0000E+00 5 -8.9765E-04 2.5621E-04 -1.2915E-04 -5.1021E-06 9.6811E-06  
 -1.2420E-06 0.0000E+00 6 -2.6288E-03 8.0824E-04 -4.4175E-05 -1.8619E-05  
 -1.2620E-05 4.5041E-06 0.0000E+00 7 -4.3474E-02 8.7969E-03 -7.7260E-04  
 -2.7259E-04 1.8367E-05 9.9215E-06 0.0000E+00 8 -1.9365E-02 1.5956E-03  
 3.4614E-04 -1.1796E-04 -1.3790E-05 5.9480E-06 -2.5281E-07  
 TABLE-US-00003 TABLE 3 # R Thickness Material CA/2 Conic coefficient K 0 Infinity  
 1.00E+06 1 3.252 2.571 K26R 2.16 -0.763 2 -7.055 0.253 1.76 -17.097 3 -10.672 0.444 EP6000  
 1.60 -75.529 4 3.302 0.309 1.45 -0.248 5 10.322 2.569 EP6000 1.47 15.386 6 -7.343 0.403 1.46  
 -43.555 7 -4.066 0.282 K26R 1.45 -22.400 8 -39.758 3.804 1.60 -20.554 9 Infinity 0.210 BK7  
 10 Infinity 0.290 11 Infinity 0.000  
 TABLE-US-00004 TABLE 4 # A1 A2 A3 A4 A5 A6 A7 1 1.6499E-03 -1.0742E-04  
 5.7901E-06 -8.6098E-08 -1.7012E-06 1.8672E-07 -2.7417E-08 2 3.0173E-03  
 -1.4633E-03 7.0329E-05 -1.5844E-05 -3.5031E-06 8.0518E-07 0.0000E+00 3  
 -6.8586E-03 5.5011E-03 -1.6856E-03 2.1537E-04 1.2470E-05 -1.0238E-05 9.8851E-07  
 4 -8.1487E-03 5.6510E-03 -7.1159E-04 1.4107E-05 3.5178E-04 1.6510E-05  
 0.0000E+00 5 -4.9793E-04 -4.5018E-04 -2.6820E-04 3.0430E-04 2.0799E-04 1.9782E-05  
 0.0000E+00 6 -2.4020E-03 1.2967E-03 -2.1528E-04 -1.8139E-04 -2.3192E-05  
 6.9007E-05 0.0000E+00 7 -6.5893E-02 1.4911E-02 -4.1874E-03 8.7863E-05 3.9488E-05  
 7.0827E-05 0.0000E+00 8 -3.4127E-02 2.0251E-03 1.8783E-03 -1.2365E-03  
 2.2451E-04 3.2977E-05 -1.1683E-05  
 TABLE-US-00005 TABLE 5 # R Thickness Material CA/2 Conic coefficient K 0 Infinity  
 1.00E+06 1 4.009 2.271 H-ZK3 2.96 0 2 18.115 1.547 2.55 0 3 -5.167 0.562 EP6000L 2.00 -2.296  
 4 6.968 0.162 2.00 9.483 5 4.666 1.082 K26R 1.90 -2.619 6 52.645 0.121 1.90 10.398 7 28.168  
 1.851 EP6000L 1.83 -367.355 8 -5.062 0.101 1.83 -10.130 9 -5.098 0.291 K26R 1.76 -10.587  
 10 15.000 4.115 1.76 -9.745 11 Infinity 0.210 BK7 2.44 12 Infinity 2.673 2.47 13 Infinity 2.94  
 TABLE-US-00006 TABLE 6 # A1 A2 A3 A4 A5 A6 A7 1 0 0 0 0 0 0 2 0 0 0 0 0 0 3  
 7.1296E-03 -1.3791E-04 -2.8926E-05 3.7349E-06 0 0 0 4 -2.8741E-03 8.8769E-04  
 -1.2786E-04 2.0275E-05 0 0 0 5 -2.1504E-03 -3.1621E-04 -3.2758E-06 -2.2831E-07 0 0 0  
 6 4.1139E-03 -1.9087E-03 1.9639E-04 -3.2249E-05 0 0 0 7 -4.3880E-03 -7.7699E-04  
 1.8992E-04 -6.8854E-06 0 0 0 8 -6.5726E-03 -5.8651E-04 1.3315E-04 -2.0025E-05 0 0 0  
 9 -7.8205E-03 -1.1425E-03 2.7014E-04 -4.0371E-05 0 0 0 10 -5.0642E-03 3.6557E-04  
 -9.7321E-05 1.7319E-05 0 0 0  
 TABLE-US-00007 TABLE 7 Material Material Conic refraction Abbe coefficient # R Thickness  
 index number CA/2 K 1 2.271 1.127 1.67 54.96 1.8 7.979E-07 2 11.822 0.06 1.725 2.410 3  
 14.756 0.27 1.64 23.52 1.7 13.805 4 2.728 0.974 1.45 2.902E-03 5 3.713 0.416 1.64 23.52 1.55  
 -2.868 6 3.524 0.764 1.5 -8.486 7 -5.301 0.338 1.64 23.52 1.48 2.743 8 -4.321 0.212 1.6  
 2.578 9 4.327 0.352 1.53 55.66 1.68 -9.755 10 3.771 2.656 1.8 -6.534 11 Infinity 0.210 1.52  
 64.16 2.894 12 Infinity 0.401 2.938 13 Infinity — 3.028  
 TABLE-US-00008 TABLE 8 # A1 A2 A3 A4 A5 A6 A7 1 4.421E-05 -2.009E-04 -1.152E-04  
 -6.051E-10 2 6.027E-03 -1.244E-03 -5.380E-08 3 0.020 7.012E-04 -1.081E-03  
 -6.297E-08 4 0.024 0.011 4.241E-04 -9.114E-08 5 -0.022 8.939E-03 2.200E-03  
 -1.002E-06 6 -0.012 6.756E-03 -2.299E-03 1.314E-03 1.758E-04 -1.030E-05 7 -0.017  
 0.053 -0.044 7.968E-03 -1.599E-03 6.117E-04 7.436E-09 8 -0.086 0.159 -0.117 0.041  
 -9.090E-03 1.280E-03 2.793E-07 9 -0.252 0.182 -0.084 0.016 -6.759E-04 -1.940E-06 10  
 -0.175 0.095 -0.040 8.597E-03 -7.751E-04 -8.160E-07

[0220] The examples provided with reference to FIGS. 2A and 2B illustrate a case where  
 CA(S.sub.1)=CH(S.sub.1). In similar cases, CA(S.sub.1) may be substantially equal to  
 CH(S.sub.1), for example with up to 5% difference.  
 [0221] In addition, an “aperture stop” **206** (which defines the lens aperture) is located before the

first surface S.sub.1. The aperture stop can be e.g. a mechanical piece. A lens module with an aperture stop located at or before the first surface S.sub.1 is known in the art as a “front aperture lens”. Lens module **200** is a front aperture lens.

[0222] Note that in other examples, the stop may be located at a different location or surface. In this case, this condition may not be true for the first surface S.sub.1. For example (this example being not limiting), the aperture stop can be located at the second surface S.sub.2. In this case,  $CA(S.sub.2)=CH(S.sub.2)$ . In similar cases  $CA(S.sub.2)$ , may be substantially equal to  $CH(S.sub.2)$ , for example with up to 5% difference.

[0223] According to some examples of the presently disclosed subject matter, there is provided an optical lens module comprising a plurality of lens elements where  $CH(S.sub.1)$  of surface S.sub.1 of lens element L.sub.1 (closest to the object side) is greater than  $CH(S.sub.k)$  of each of all other surfaces S.sub.k of the plurality of lens elements, with  $2 \leq k \leq 2N$ .

[0224] For example, if  $N=4$  (as in FIGS. 2A, 2B and 6),  $CH(S.sub.1)$  is greater than  $CH(S.sub.2)$ ,  $CH(S.sub.3)$ ,  $CH(S.sub.4)$ ,  $CH(S.sub.5)$ ,  $CH(S.sub.6)$ ,  $CH(S.sub.7)$  and  $CH(S.sub.8)$ . This applies to different values of N.

[0225] For example, if  $N=4$  (as in FIGS. 2A, 2B and 6),  $CH(S.sub.2)$  is greater than  $CH(S.sub.3)$ ,  $CH(S.sub.4)$ ,  $CH(S.sub.5)$ ,  $CH(S.sub.6)$ ,  $CH(S.sub.7)$  and  $CH(S.sub.8)$ . This applies to different values of N.

[0226] For example, if  $N=5$  (as in FIG. 20),  $CH(S.sub.1)$  is greater than  $CH(S.sub.2)$ ,  $CH(S.sub.3)$ ,  $CH(S.sub.4)$ ,  $CH(S.sub.5)$ ,  $CH(S.sub.6)$ ,  $CH(S.sub.7)$ ,  $CH(S.sub.8)$ ,  $CH(S.sub.9)$  and  $CH(S.sub.10)$ . This applies to different values of N.

[0227] For example, if  $N=5$  (as in FIG. 20),  $CH(S.sub.2)$  is greater than  $CH(S.sub.3)$ ,  $CH(S.sub.4)$ ,  $CH(S.sub.5)$ ,  $CH(S.sub.6)$ ,  $CH(S.sub.7)$ ,  $CH(S.sub.8)$ ,  $CH(S.sub.9)$  and  $CH(S.sub.10)$ . This applies to different values of N.

[0228] According to some examples,  $CH(S.sub.1) \geq X \times CH(S.sub.2)$ , wherein X is any value in the range [1.01;2] (such as  $X=1.1$  or any other value in this range).

[0229] According to some examples, the following conditions are fulfilled: [0230]  $CH(S.sub.1) \geq 1.1 \times CH(S.sub.2)$ , and [0231]  $CH(S.sub.1) \geq 1.2 \times CH(S.sub.k)$ , for each of all other surfaces S.sub.k, with  $3 \leq k \leq 2N$ .

[0232] According to some examples, the following conditions are fulfilled: [0233]  $CH(S.sub.1) \geq 1.45 \times CH(S.sub.k)$ , for each of all other surfaces S.sub.k, with  $3 \leq k \leq 2N$ .

[0234] According to some examples, the following condition is fulfilled: [0235]  $CH(S.sub.2) \geq 1.1 \times CH(S.sub.k)$ , for each of surfaces S.sub.k, with  $3 \leq k \leq 2N$ .

[0236] According to some examples, the following conditions are fulfilled: [0237]  $CH(S.sub.1) \geq X \times CH(S.sub.2)$ , and [0238]  $CH(S.sub.1) \geq Y \times CH(S.sub.k)$ , for each of all other surfaces S.sub.k, with  $3 \leq k \leq 2N$ , where  $Y > X$ . In some examples, X can be any value in the range [1.01;2], and Y can be any value in the range [1.01;2].

[0239] According to some examples, the following conditions are fulfilled: [0240]  $CH(S.sub.2) \geq Y \times CH(S.sub.k)$ , for each of all other surfaces S.sub.k, with  $3 \leq k \leq 2N$ , where  $Y > X$ . In some examples, Y can be any value in the range [1.01;2].

[0241] According to some examples,  $CA(S.sub.1)$  of surface S.sub.1 of lens element L.sub.1 is greater than  $CA(S.sub.k)$  of each of all other surfaces S.sub.k of the plurality of lens elements, with  $2 \leq k \leq 2N$ . According to some examples,  $CA(S.sub.2)$  of surface S.sub.2 of lens element L.sub.1 is greater than  $CA(S.sub.k)$  with  $3 \leq k \leq 2N$ .

[0242] For example, if  $N=4$  (as in FIGS. 2A and 2B),  $CA(S.sub.1)$  is greater than  $CA(S.sub.2)$ ,  $CA(S.sub.3)$ ,  $CA(S.sub.4)$ ,  $CA(S.sub.5)$ ,  $CA(S.sub.6)$ ,  $CA(S.sub.7)$  and  $CA(S.sub.8)$ . This applies to different values of N.

[0243] According to some examples,  $CA(S.sub.1) \geq X \times CA(S.sub.2)$ , wherein X is any value in the range [1.01;2] (such as  $X=1.1$  or any other value in this range).

[0244] According to some examples, the following conditions are fulfilled: [0245]

CA(S.sub.1)≥1.1×CA(S.sub.2), and [0246] CA(S.sub.1)≥1.2×CA(S.sub.k), for each of all other surfaces S.sub.k, with 3≤k≤2N.

[0247] According to some examples, the following conditions are fulfilled: [0248] CA(S.sub.1)≥1.45×CA(S.sub.k), for each of all other surfaces S.sub.k, with 3≤k≤2N.

[0249] According to some examples, the following condition is fulfilled: [0250] CA(S.sub.2)≥1.1×CA(S.sub.k), for each of surfaces S.sub.k, with 3≤k≤2N.

[0251] According to some examples, the following conditions are fulfilled: [0252] Y×CA(S.sub.1)≥X×CA(S.sub.2), and [0253] CA(S.sub.1)≥Y×CA(S.sub.k), for each of all other surfaces S.sub.k, with 3≤k≤2N,

where Y>X. In some examples, X can be any value in the range [1.01;2], and Y can be any value in the range [1.01;2].

[0254] According to some examples, CA(S.sub.1) is substantially equal to CA(S.sub.2N) and is greater than CA(S.sub.k) of each of all other surfaces S.sub.k of the plurality of lens elements, with 2≤k≤2N-1. For example, if N=5 (as in FIGS. 21A and 21B), CA(S.sub.1)=CA(S.sub.10) is greater than CA(S.sub.2), CA(S.sub.3), CA(S.sub.4), CA(S.sub.5), CA(S.sub.6), CA(S.sub.7), CA(S.sub.8) and CA(S.sub.9). This applies to different values of N. In similar cases CA(S.sub.1) may be substantially equal to CA(S.sub.10), for example with up to 5% difference.

[0255] According to some examples, the following conditions are fulfilled: [0256] CA(S.sub.1)≥1.05×CA(S.sub.2), and [0257] CA(S.sub.1)≥1.1×CA(S.sub.k), for each of all other surfaces S.sub.k, with 3≤k≤2N-1.

[0258] According to some examples, the following condition is fulfilled:  
[00002]BFL ≥ X × TTL,

In this equation, X is any value in the range [0.2;0.5]. According to some examples, X=0.3 or X=0.4 where TTL and BFL are defined above.

[0259] In FIGS. 2A and 2B, the BFL is measured between the center of the surface S.sub.8 and the image sensor 202.

[0260] In FIGS. 2A and 2B, the TTL is measured between the center of the surface S.sub.k and the image sensor 202.

[0261] This configuration of the relative values of BFL and TTL as disclosed above can improve the quality of the image formed on the image sensor.

[0262] Using a lens element L.sub.1 with a front surface that has a greater CH value or CA value with respect to the other surfaces can help to increase the amount of incoming light which can be sensed by the image sensor of the camera or of the Tele sub-camera.

[0263] Advantageously, f/# (f-number) can be less than 3.

[0264] Advantageously, S.sub.1 and/or S.sub.2 can be spherical.

[0265] Advantageously, the ratio between the last lens element clear aperture CA(S.sub.2N) and the sensor diagonal (SD) may be less than 0.8 or 0.7 or 0.65.

[0266] Advantageously, TTL may be smaller than EFL.

[0267] According to some examples of the presently disclosed subject matter (tables 1-4), all lens elements L.sub.1 to L.sub.N may be made of plastic material. According to some examples of the presently disclosed subject matter (Tables 5-6), lens element L.sub.1 may be made of glass material and all lens elements L.sub.2 to L.sub.N may be made of plastic material. This is however non-limiting and lens elements L.sub.1 to L.sub.N may all be made by either plastic or glass material. The selection of lens element material (plastic or glass) is influenced by various optical and mechanical demands. For example, and as known in the art, different materials (glass and/or plastic) have different refractive indexes, glass having typically a higher refractive index selection range than plastic. For example, different materials have different Abbe numbers, glass having typically a higher Abbe number selection range than plastic. An example for 3 materials, refractive indexes and Abbe numbers is given above, out of hundreds of materials with corresponding Abbe numbers and refractive indexes available. For example, the surface profiles of plastic lens elements



may be approximated by a polynomial with many coefficients (4-7 in the examples in Tables 1-6), while surface profiles of glass lens elements can be approximated in a similar way when molded or may be limited to spherical shape when polished (0 coefficient in the examples in Tables 5-6). This limitation is driven from manufacturing limits known in the art. For example, the minimal thickness of a glass lens element may be smaller than that of a plastic element, as known in the art. For example, glass lens elements may be cut (or diced or sliced) to a non-circular shape, as demonstrated in FIGS. **13A-13C**.

[0268] In addition to the fact that at least the first lens element can be of increased dimensions in order to increase light impinging on the sensor, according to some examples, the barrel that holds the lens elements has to be mechanically resilient to external stress while striving to maintain the module height (along an axis perpendicular to the optical axis of barrel, which corresponds to axis Y in the figures) as low as possible. This is advantageous for example when it is desired to fit a camera within the limited available space (e.g. thickness) of a computerized device such as a Smartphone.

[0269] Examples of an optical lens module which is designed to deal with these contradictory requirements are described with reference to FIGS. **6-11, 13, 17-19** and **22-25**. The optical lens module is not limited to the examples described with reference to FIGS. **6-11, 13, 17-19** and **22-25**.

[0270] In the example illustrated in FIG. **6**, barrel **64** of optical lens module **60** comprises a cavity **61** surrounded by walls **62**. In this example, a first subset of the lens elements is held in cavity **61**, and a second subset of the lens elements is located externally to barrel.

[0271] In particular, according to the example shown in FIG. **6**, lens elements L.sub.2 to L.sub.N are held within cavity **61** and lens element L.sub.1 is located outside of barrel **64** (that is to say that lens element L.sub.1 is not within cavity **61**). Lens element L.sub.1 can be affixed to barrel **64**, by any appropriate mechanical link, such as an adhesive material.

[0272] In other examples, lens elements L.sub.1 to L.sub.i (with  $1 < i < N$ ) are located outside of barrel **64** (out of cavity **61**), and lens elements L.sub.i to L.sub.N are held within cavity **61**.

[0273] In the example of FIG. **6**, since lens element L.sub.1 is located outside of cavity **61**, the height H.sub.L1 of lens element L.sub.1 can be substantially equal to or larger than the height H of barrel **64** (measured along the axis Y between external surfaces of opposite walls of barrel **64**). Heights H.sub.L2 to H.sub.LN of lens element L.sub.2 to L.sub.N can be smaller than the height H of barrel **64**. A numerical (non-limiting) example for lens **60** may be have the following values: H.sub.L1=4.82 mm, H.sub.L2=H.sub.L3=H.sub.L4=3.7 mm.

[0274] Attention is now drawn to FIG. **7**, which depicts another example of an optical lens module **70**.

[0275] In this example, optical lens module **70** comprises a barrel **74**. Barrel **74** comprises a cavity **71** circumvented by walls **72**. According to the example illustrated in FIG. **7**, lens elements L.sub.1 to L.sub.N can be all located within cavity **71**.

[0276] According to some examples, a height H.sub.C of cavity **71**, measured along an axis orthogonal to optical axis **103** (between internal parts **73**), is variable along optical axis **103**.

[0277] In the representation of FIG. **7**, height H.sub.C of cavity **71** is measured along axis Y. For each position along the Z axis, height H.sub.C corresponds in this example to the distance between the internal parts **73** of walls **72** along axis Y. For cases in which the cavity barrel is axial symmetric, H.sub.C is the internal diameter of the cavity barrel. In the example of FIG. **7**, cavity height H.sub.C is variable along axis Z. In other words, H.sub.C(Z) is not a constant function.

[0278] According to some examples, cavity **71** comprises a first portion **76** in which first lens element L.sub.1 is located and a second portion **77** in which at least some of the other lens elements (L.sub.2 to L.sub.N) are located.

[0279] According to this example, height H.sub.C(Z.sub.1) of first portion **76** of cavity **71** is greater than height H.sub.C(Z.sub.2) of second portion **77** of cavity **71**. As a consequence, first lens element L.sub.1 (which is generally of greater dimensions, as mentioned above) is positioned

within first portion **76** of cavity **71**, and at least some of the other lens elements are positioned within second portion **77** of cavity **71**.

[0280] According to this example, height  $H_{\text{sub.C}}(Z_{\text{sub.1}})$  of first portion **76** of cavity **77** is designed to fit with the height  $H_{\text{sub.L1}}$  of first lens element  $L_{\text{sub.1}}$ , and height  $H_{\text{sub.C}}(Z_2)$  of the second portion **77** of cavity **71** is designed to fit with height  $H_{\text{sub.L2}}$ ,  $H_{\text{sub.L3}}$  and  $H_{\text{sub.L4}}$  of the other lens element  $L_{\text{sub.2}}$  to  $L_{\text{sub.4}}$  (in this example,  $H_{\text{sub.L2}}=H_{\text{sub.L3}}=H_{\text{sub.L4}}$ ).

[0281] The variable height of cavity **71** along optical axis **103** can be obtained e.g. by using walls **72** with a variable thickness. As shown in FIG. 7, walls **72** have a thinner thickness in first portion **76** than in second portion **77**. In other examples, walls **72** have a constant thickness but they have a stepped shape.

[0282] Various examples (see FIGS. **8** to **13** and **17-19**) of an optical lens module comprising a plurality of lens elements  $L_{\text{sub.1}}$  to  $L_{\text{sub.N}}$  will now be described. These optical lens modules can be used in any of the examples of cameras or of optical designs described above. In these examples (see FIGS. **8** to **13** and **17-19**), the relationship between the dimensions of the lens elements can be in accordance with any of the examples described above (see e.g. FIG. 2A to FIG. 5 and Tables 1-6) and are thus not described again.

[0283] According to some examples, the height of lens element  $L_{\text{sub.1}}$  is greater than the height each of lens elements  $L_{\text{sub.2}}$  to  $L_{\text{sub.N}}$  (in the examples of FIGS. **8** to **13** and **17-19**). Other relationships can be defined, as already explained above (these definitions can rely e.g. on a relationship between the clear apertures and/or the clear heights of the lens elements).

[0284] Attention is now drawn to FIG. **8**, which depicts an example of an optical lens module **80** comprising a plurality of lens elements  $L_{\text{sub.1}}$  to  $L_{\text{sub.N}}$ . In this example, four lens elements  $L_{\text{sub.1}}$  to  $L_{\text{sub.4}}$  are depicted. In this example, optical lens module **80** comprises a barrel **84**. Barrel comprises a cavity **81** circumvented by walls **82**. At least some of lens elements  $L_{\text{sub.2}}$  to  $L_{\text{sub.N}}$  are located within cavity **81**.

[0285] The lens elements which are within cavity **81** have a center which is substantially aligned with optical axis **103**. The center of a lens element can be defined as the physical center of the whole lens element (including the optical part and the mechanical part of the lens element, see e.g. in FIG. 11F wherein the physical center can be located at the center of the total height  $H_{\text{sub.L}}$  of the lens element), or as the center of only the optical portion of the lens element (see e.g. in FIG. 11F wherein the optical center can be located at the center of the optical height  $H_{\text{sub.opt}}$  of the lens element). Generally, the physical center of the lens element coincides with the center of the optical part (optical center) of the lens element. For an axial symmetric lens element,  $H_{\text{sub.opt}}$  is defined as the maximum of clear apertures of front and back surfaces of the respective lens element.

[0286] In this example, extremity **83** of walls **82** is shaped so that extremity **83** of walls **82** acts a stop for at least a portion of lens element  $L_{\text{sub.1}}$ .

[0287] In particular, lens element  $L_{\text{sub.1}}$  is prevented from moving in the Y-Z plane by extremity **83** of the walls acting as a mechanical stop. By choosing an appropriate shape and appropriate dimensions for extremity **83** of the walls **82**, and likewise shaping a part of lens element  $L_{\text{sub.1}}$  to fit the shape of extremity **83**, the center of lens element  $L_{\text{sub.1}}$  can be substantially aligned with optical axis **103**.

[0288] In the example of FIG. **8**, the cross-section of extremity **83** of walls **82** has a stepped shape.

[0289] An extremal portion **85** (this portion is part of the thickness of the lens element) of lens element  $L_{\text{sub.1}}$  is located within cavity **81**. In some examples, extremal portion **85** corresponds to the rear surface of lens element  $L_{\text{sub.1}}$ .

[0290] A main portion **86** (this portion is part of the thickness of the lens element) of lens element  $L_{\text{sub.1}}$  is located outside of cavity **81**. In some examples, a thickness of extremal portion **85** measured along optical axis **103** is less than a thickness of main portion **86** measured along optical axis **103**. Extremal portion **85** of lens element  $L_{\text{sub.1}}$  is blocked between walls **82**. In particular,

the stepped shape of extremity **83** of walls **82** is made to match or to fit a part **87** of extremal portion **85** of lens element L.sub.1, wherein part **87** has also a stepped shape in cross-section. As apparent in FIG. **8**, the stepped shape of extremity **83** fits together with the stepped shape of part **87** of lens element L.sub.1. Therefore, lens element L.sub.1 is prevented from moving in the Y-Z plane, and the center of lens element L.sub.1 can be maintained in alignment with optical axis **103**. [0291] FIG. **9A** describes another example of an optical lens module. In this example, the cross-section of extremity **93** of walls **92** has a sloped shape. In particular, extremity **93** can be shaped as a chamfer. An extremal portion **95** (this portion is considered in the width of the lens element) of lens element L.sub.1 is located within a cavity **91** of barrel **94** of optical lens module **90**. In some examples, extremal portion **95** corresponds to the rear surface of lens element L.sub.1. A main portion **96** (this portion is part of the thickness of the lens element) of lens element L.sub.1 is located outside of cavity **91**. Extremal portion **95** of lens element L.sub.1 is prevented from moving in the Y-Z plane by extremity **93** of walls **92**.

[0292] In particular, the sloping shape of extremity **93** of walls **92** is made to match or to fit a part **97** of extremal portion **95** of lens element L.sub.1, wherein part **97** has also a sloping shape in cross-section. As apparent from FIG. **9A**, the sloping shape of extremity **93** fits together with the sloping shape of part **97**. Therefore, lens element L.sub.1 is prevented from moving in the Y-Z plane, and the center of lens element L.sub.1 can be maintained in alignment with optical axis **103**. [0293] FIG. **9B** depicts a variant of FIG. **9A**. In this example, a portion **98** of lens element L.sub.1 is located within cavity. This portion **98** can correspond to a main portion of lens element L.sub.1 or to the whole lens element L.sub.1. Extremity **93** of walls **92** has a sloping shape in cross-section, as in the example of FIG. **9A**, but in this example the slope extends further along the side of lens element L.sub.1. A part **97** of portion **98** is in contact with extremity **93** of walls **92**, and has also a sloping shape in cross-section, which fits together with the sloping shape of extremity **93**.

Therefore, portion **98** of lens element L.sub.1 is prevented from moving in the Y-Z plane.

[0294] FIG. **10** describes another example. In this example, an extremal portion **1005** (this portion is part of the width of the lens element) of lens element L.sub.1 is located within a cavity **1001**, whereas a main portion **1006** (this portion is part of the width of the lens element) of lens element L.sub.1 is located outside of cavity **1001**. In some examples, extremal portion **1005** corresponds to the rear surface of lens element L.sub.1.

[0295] In some examples, a thickness of the extremal portion **1005** measured along optical axis **103** is less than a thickness of main portion **1006** measured along optical axis **103**.

[0296] Extremal portion **1005** of lens element L.sub.1 is blocked between walls **1002**. In particular, a part **1007** of extremal portion **1005** which is in contact with an extremity **1003** of walls **1002** has a stepped shape. Extremity **1003** of walls **1002** acts as a stop for extremal portion **1005**, since part **1007** is blocked by extremity **1003** and is prevented from moving in the Y-Z plane. Therefore, lens element L.sub.1 is prevented from moving in the Y-Z plane, and the center of lens element L.sub.1 can be maintained in alignment with optical axis **103**.

[0297] In this example, the shape of walls **1002** can be uniform. In particular, the shape of extremity **1003** of walls **1002** can be identical with the shape of the other portions of walls **1002**, contrary to the examples described in FIGS. **8**, **9** and **9A**, and only a part of the lens element is needed to be shaped in order to fit extremity **1003**.

[0298] According to some variants of the example of FIG. **10**, a main portion of lens element L.sub.1 is located within the cavity (and not only an extremal portion as in FIG. **10**) and the extremity (see reference **1009** in FIG. **10**) of the walls matches a part of lens element L.sub.1 which has a stepped shape. Lens element L.sub.1 is thus prevented from moving in the Y-Z plane.

[0299] Attention is now drawn to FIGS. **11A** to **11E**.

[0300] According to some examples, an optical lens module **1100** can comprise lens elements L.sub.1 to L.sub.N and a barrel **1114**. Barrel **1114** comprises a cavity **1101** surrounded by walls **1102**. N lens elements L.sub.1 to L.sub.N are located within cavity **1101**. In this example, N is

equal to four. The optical lens module can further comprise stops **1115**, which can be present between each of two adjacent lens elements. Stops **1115** can have an annular shape. These stops **1115** are useful for maintaining the lens elements at their required position and for maintaining the required distance between the lens elements.

[0301] A height of barrel **1114**, which can be measured, for example, between external surfaces **1103** of opposite walls **1104** of barrel **1114** (e.g. along an axis Y orthogonal to a symmetry axis of barrel **1114**) is equal to H. In the examples of FIGS. **11A** to **11E**, the height H.sub.L1 of lens element L.sub.1 can be substantially equal to H or greater than H. Thus, lens element L.sub.1 can have a large height (thus benefiting from an increased optical collection surface) while being located within the optical lens module which provides protection and mechanical support of lens element L.sub.1. With this configuration, the center of lens element L.sub.1 can be maintained in alignment with optical axis **103**.

[0302] In addition, a lens element generally has an optical part and a mechanical part. The mechanical part is the part of the lens element which is not used for transmitting rays. This is visible for example in FIG. **11A**, in which lens element L.sub.2 comprises an optical part **1130** and a mechanical part **1131**. This is also illustrated in FIG. **11F**.

[0303] According to some examples, the ratio between a height of the optical part (see H.sub.opt in FIG. **11F**) and the height of the lens element (see H.sub.L in FIG. **11F**) is greater for lens element L.sub.1 than for each of lens elements L.sub.2 to L.sub.N.

[0304] As shown in the figures, barrel **1114** can comprise slots **1110** on two of the opposite walls **1111** of barrel **1114**. This allows the lens element L.sub.1 to be substantially of the same height as the barrel, or to have a height which is greater than the barrel, and to have a height which is greater than that of the other lens elements. In particular, lens element L.sub.1 can be tangent to slots **1110**, or at least part of the lens element L.sub.1 can protrude through slots **1110**.

[0305] Attention is now drawn to FIG. **12** which describes an example of a manufacturing method of the optical lens module of FIGS. **11A** to **11E**. The method can comprise a step **1200** of providing a barrel comprising walls defining a cavity. The barrel can already comprise slots on at least two opposite walls. Alternatively, the method can comprise creating slots in at least two opposite walls of the barrel.

[0306] The method can comprise a step **1201** of inserting lens elements L.sub.1 to L.sub.N in the cavity of the barrel. Generally, lens element L.sub.N, which is the closest to the image side, is the first lens element to be inserted. Lens element L.sub.N can be fastened to the barrel using a fastening material such as an adhesive, so that it acts as a stop at one side of cavity for the other lens elements.

[0307] According to some examples, stops are inserted within cavity, between the lens elements, as already discussed with respect to FIG. **11A**. The lens elements are thus stacked within cavity. In order to prevent the lens elements from moving from their position, the method can comprise a step **1202** of fastening at least lens element L.sub.1 in order to maintain lens elements L.sub.1 to L.sub.N within cavity. This can be performed by injecting a fastening material (e.g. adhesive) within the cavity, e.g. through adapted through-holes **1120** present in the walls of the barrel. The adhesive thus fastens lens element L.sub.1 to the internal surfaces of the walls. After these steps, and if necessary, the through holes can then be plugged.

[0308] The structure of the lens module as depicted in FIGS. **11A** to **11E** is thus also advantageous in terms of the manufacturing process, since lens element L.sub.1 can be of the height of barrel (or can have a height which is greater than the height of the barrel) and can still be fastened to the internal surfaces of the walls of the barrel (and not as in FIG. **6**, where lens element L.sub.1 is outside of the barrel and affixed only to an extremity of the walls of the barrel).

[0309] Attention is now drawn to FIG. **13A**, which depicts an example of an optical lens module **1380** comprising a plurality of lens elements **1381**. Optical lens module **1380** comprises a barrel **1314**. At least some of the lens elements can be located within barrel **1314**.

[0310] According to some examples, at least part of the lens elements can have a shape (profile) in cross-section (in plane X-Y, which is orthogonal to the optical lens module and which generally coincides with the optical axis) which is not circular. In particular, as shown e.g. in FIG. 13A, at least some of the lens elements can have a width  $W_{\text{sub.L}}$  (measured along axis X) which is greater than their height  $H_{\text{sub.L}}$  (measured along axis Y). The height  $H_{\text{sub.L}}$  can correspond to the total height of the lens element (including the mechanical part). In some embodiments, a lens element in lens module 1380 may have a symmetry about axis Y and/or about axis X.

[0311] According to some examples,  $W_{\text{sub.L}}$  is substantially greater than  $H_{\text{sub.L}}$  (for example, by at least a percentage which is equal or greater than 10%, these values being not limiting).

[0312] According to some examples, at least part of the lens elements is shaped so as their profile in cross-section comprises sides with straight portions. Other sides of the profile can be e.g. curved. This can be seen e.g. in FIG. 13A, wherein sides 1350 (in this example two sides) of the profile of these lens elements in cross-section are substantially straight lines along axis X. As a consequence, at least some of the sides of these lens elements are flat surfaces (in plane X-Z). In FIG. 13A, the two other sides 1351 of the profile of these lens elements in cross-section are curved lines.

[0313] According to some examples, barrel 1314 is shaped to fit with the shape of the lens elements. Thus, barrel 1314 can have walls which have a profile in cross-section which is similar to the profile of the lens elements (located in barrel) in cross-section.

[0314] It is to be noted that other shapes and profiles can be used for the lens elements (and thus for barrel), such as (but not limited to) an elliptic profile.

[0315] The configuration described with reference to FIGS. 13A to 13C allows in particular to increase the quantity of light received by the image sensor, for a given height of barrel.

[0316] In the example depicted in FIG. 13B, lens element L.sub.1, which is the closest lens element to the object side, is located outside of barrel 1314. Examples wherein lens element L.sub.1 is positioned outside of barrel have been described e.g. with reference to FIG. 6, and examples wherein a main portion (measured along the thickness of the lens element) of lens element L.sub.1 is located outside of barrel 1314 have been described above (see e.g. FIGS. 8 to 11E). At least part of the features described with reference to these examples can be used in the example of FIG. 13B and are not described again.

[0317] In the example depicted in FIG. 13C, lens element L.sub.1, which is the closest lens element to the object side, is also located within barrel 1314. Examples wherein lens element L.sub.1 is positioned within the barrel have been described e.g. with reference to FIG. 7, and examples wherein a main portion (measured along the thickness of the lens element) of lens element L.sub.1 is located within the barrel have been described above (see e.g. the description of FIGS. 8, 9A and 10). At least part of the features described with reference to these examples can be used in the examples of FIG. 13A and FIG. 13C and are not described again.

[0318] Attention is now drawn to FIGS. 17A to 17D. FIG. 17A shows an isometric view of a lens module 1700. FIG. 17B shows a side view of lens module 1700. FIG. 17C shows an exploded view of lens 1700. Lens module 1700 may have an optical design similar to lens 200. Lens module 1700 includes a barrel 1720. Lens module 1700 further includes lens elements L.sub.1 to L.sub.N. N is normally in the range of 3-7, similar to lens module 200. In the non-limiting example of lens module 1700, N=4. Lens module 1700 has the first lens element L.sub.1 partially positioned or placed outside of barrel 1720, while lens elements L.sub.2 to L.sub.N are placed completely inside the barrel. L.sub.1 is clearly seen in FIG. 17A, while other lens elements are not seen in this view but can be seen in FIG. 17B. Lens module 1700 has an optical axis 103 which serves as an axial symmetry axis for all lens element L.sub.1 to L.sub.N. Each lens element L.sub.i has a height  $H_{\text{sub.Li}}$  defined along the Y axis. Lens element L.sub.1 may have a “stepped” shape, i.e. it has a front part with height  $H_{\text{sub.L1}}$  and a back part with height  $H_{\text{sub.L1B}}$ , such that  $H_{\text{sub.L1}} > H_{\text{sub.L1B}}$ . Lens module 1700 may further include spacers R.sub.1 to R.sub.N-1. Each spacer R.sub.i is positioned between lens elements L.sub.i and L.sub.i+1. In some embodiments,

one or more of spacers  $R_{sub.1}$  to  $R_{sub.N-1}$  may be used as an aperture stop(s).

[0319] Barrel **1720** may be made for example from opaque plastic using plastic injection molding, as known in the art. Barrel **1720** has a cavity **1740** that may be axial-symmetric along optical axis **103**. Cavity **1740** may have a shape of cylinder as in embodiment **1700** (FIG. **17B**). In other embodiments, cavity **1740** may have other axial symmetric shapes, such as a cone, a series of cylinders, etc. (see FIG. **17D**). The axial symmetry accuracy of cavity **1740** is important for the assembling accuracy, as known in the art. In some embodiments, the tolerance for axial symmetry distortion may be smaller than 10  $\mu\text{m}$ , 5  $\mu\text{m}$  or 1  $\mu\text{m}$ .

[0320] Lens elements  $L_{sub.i}$  may be made by plastic injection molding, as known in the art. Lens elements  $L_{sub.i}$  may be made from glass, as known in the art. Each lens element  $L_{sub.i}$  has front surface ( $S_{sub.2i-1}$ ) and back surface ( $S_{sub.2i}$ ) as defined above for embodiment **200**. Each surface  $S_{sub.k}$  ( $3 \leq k \leq 2N$ ) may have an optically active part and a mechanical part which is a non-active optical part (as described in FIG. **11F**). A mechanical part may be used to handle the lens element during the molding and assembly stages. In some examples the mechanical part size may be on the order of 100  $\mu\text{m}$ -500  $\mu\text{m}$ . For example, the mechanical part of  $S_{sub.1}$  is marked with numeral **1751**. The closest point of  $L_{sub.1}$  to the object side is marked with numeral **1752**.

[0321] FIG. **17D** shows a lens module **1701** which is similar to lens module **1700** with a single difference: barrel **1722** with cavity **1742** replaces barrel **1720** with cavity **1740**. Cavity **1742** has a shape of a series of cylinders in increasing sizes for each lens element; as can be seen in FIG. **17D**,  $H_{sub.L1B} \leq H_{sub.L2} \leq H_{sub.L3} \leq H_{sub.L4} < H_{sub.L1}$ . This feature may allow easier molding of barrel **1720** and/or easier assembly of lens elements  $L_{sub.1}$  to  $L_{sub.4}$  and spacers  $R_{sub.1}$  to  $R_{sub.3}$ . In other embodiment the number of lens elements may differ from 4 as mentioned above. A numerical (non-limiting) example for lens module **1700** may be have the following values:  $H_{sub.L1} = 4.9 \text{ mm}$ ,  $H_{sub.L1B} = 3.65 \text{ mm}$ ,  $H_{sub.L2} = 3.7 \text{ mm}$ ,  $H_{sub.L3} = 3.8 \text{ mm}$ ,  $H_{sub.L4} = 3.9 \text{ mm}$ .

[0322] The assembly of lens module **1700** (or **1701**) may be done in the following steps: [0323] 1. Insertion of lens element  $L_{sub.1}$  from the object side of barrel **1720**.  $L_{sub.1}$  may be aligned to barrel **1720** due to the axial symmetry of both elements. [0324] 2. Gluing of  $L_{sub.1}$  to barrel **1720**. Gluing may be done using glue on surface **1722**, which is the front-most surface of barrel **1722**. [0325] 3. Insertion of other elements from the image side of barrel in the following order:  $R_{sub.1}$ ,  $L_{sub.2}$  . . .  $R_{sub.N-1}$ ,  $L_{sub.N}$ .  $L_{sub.2}$  to  $L_{sub.N}$  and  $R_{sub.1}$  to  $R_{sub.N-1}$  may be aligned to barrel **1720** due to the axial symmetry of all elements. [0326] 4. Gluing lens element  $L_{sub.N}$  to barrel **1720**, for a non-limiting example on surface **1724** which is the inner surface of barrel **1722**. [0327] Attention is now drawn to FIGS. **18A** to **18C**. FIG. **18A** shows an isometric view of lens module **1800** which is similar to lens module **1700**, except that it has an added cover **1830**. All other parts (barrel, lens elements, optical axis) are as in lens module **1700** and have the same numbering and names. FIG. **18B** shows a side cut of lens module **1800**. FIG. **18C** shows an exploded view of lens module **1800**. Cover **1830** may be made from opaque plastic, e.g. by plastic injection molding. Cover **1830** is positioned on top of lens element  $L_{sub.1}$ . In some embodiments, cover **1830** may optically cover mechanical part **1751** such that cover **1830** prevents any optical ray of light arriving from the OPFE from reaching mechanical part **1751**. In some embodiments, cover **1830** may have a point **1831** which is closer to the object than point **1752** on  $L_{sub.1}$ . This feature is important in order to protect lens module **1800** while handling and assembling, such that the risk of having lens element  $L_{sub.1}$  touching accidentally another object is minimized.

[0328] The assembly process of lens module **1800** may be similar to the assembly process of lens module **1700** above with an addition of a fifth step: [0329] 5. Positioning of cover **1830** and gluing it to barrel **1720** or to  $L_{sub.1}$ . In one example gluing may be done on surface **1724**.

[0330] Attention is now drawn to FIGS. **19A** and **19B**, which show a lens module **1900** similar to lens module **1800**, except that a barrel **1920** replaces barrel **1720**. The change to barrel **1920** allows a different assembly process (relative to lens module **1800**), detailed below. FIG. **19A** shows a side

cut of lens **1900** and FIG. **19B** shows lens module **1900** in an exploded view, according to the different assembly direction.

[0331] The assembly of lens module **1900** may be done in the following steps: [0332] 1. Insertion of lens element L.sub.N from the object side of barrel **1820**. [0333] 2. Insertion of other elements from the object side of barrel in the following order: R.sub.N-1, L.sub.N-1, . . . R.sub.1, L.sub.1 [0334] 3. Gluing lens element L.sub.1 to barrel **1820** for a non-limiting example on surface **1724**. [0335] 4. Positioning of cover **1730** and gluing it to barrel **1820** or to L.sub.1. In one example gluing may be done on surface **1724**.

[0336] The presently disclosed subject matter also contemplates a method of forming an image on an image sensor, using any of the examples described above.

[0337] The presently disclosed subject matter also contemplates a method of producing an optical lens module according to the specifications as described by any of the examples above.

[0338] According to some examples, the digital camera can be integrated inside a housing of a mobile electronic device (such as, but not limited to, a smartphone, a portable computer, a watch, eyewear, etc.).

[0339] According to some examples, the optical lens module, associated with the lens elements, (as described in the various examples above), can be integrated in a digital camera, or in a Tele sub-camera or in a plurality of Tele sub-cameras of a digital camera. This digital camera can in addition comprise one or more Wide sub-cameras.

[0340] A folded camera can be used to reduce the height of elements of the camera. As mentioned above, this can e.g. facilitate the integration of the camera when only limited space is available.

[0341] According to at least some of the examples described above, the proposed solution can increase image quality by increasing the incoming light through the camera aperture. This can be achieved notwithstanding an increase of the distance (along Z axis) between the first lens element (at the object side) and the image sensor, as a result of a longer EFL used for obtaining an increased zoom factor.

[0342] In addition, according to at least some of the examples described above, the proposed solution can offer an optical lens module which can firmly hold the lens elements while complying with the limited available height.

[0343] In addition, according to at least some of the examples described above, the quantity of light which is collected by the sensor is increased for a given height of barrel of the optical lens module.

[0344] As explained above, using a lens element L.sub.1 incorporated in a lens module (the lens module comprising a plurality of lens elements, each having a front surface and a rear surface) with a front surface that has a greater CH (clear height) value or greater CA (clear aperture) value with respect to the other surfaces helps to increase the incoming light which enters the lenses barrel and can be sensed by an image sensor of the camera (e.g. Tele sub-camera in a dual aperture camera). As more light can reach the sensor such configuration enables to increase the focal length of the lens module.

[0345] It is known that a negative correlation exists between the focal length and a respective field of view, where the field of view becomes smaller as the focal length increases. Thus, while an increase to a given focal length in a given camera enables to increase image resolution, the higher resolution image is formed on a smaller area of the camera sensor. In other words, when capturing an image of the same object from the same distance with two lenses, one having a focal length longer the other, the lens module with the longer focal length produces on the sensor a smaller image with higher spatial resolution as compared to the one with the shorter focal length. Thus, the advantages of a larger focal length are accompanied with the disadvantage of a smaller size image.

[0346] Accordingly, some examples of the presently disclosed subject matter include a digital camera as disclosed above comprising:

[0347] N lens elements L.sub.i (lens module) having a symmetry along a first optical axis, each lens element comprising a respective front surface S.sub.2i-1 and a respective rear surface

S.sub.2i, where i is between 1 and N, and N is equal to or greater than 3; wherein a clear height value of surface S.sub.1 or a clear aperture value of surface S1 is greater than a clear height value or a clear aperture value of each of surfaces S.sub.2 to S.sub.2N;

[0348] The digital camera further comprises an image sensor and a rotating reflecting element or OPFE (such as a mirror or prism). The rotating reflecting element is inclined with respect to the first optical axis, so as to provide a folded optical path between an object and the lens elements and is capable of being rotated around one or two axes.

[0349] An example of such rotating reflecting element is disclosed, by way of example in co-owned international patent application PCT/IB2017/052383, which describes an actuator of a digital camera designed to enable the rotation of a reflecting element around two axes. See for example FIG. 1A to FIG. 1F, FIG. 2 and FIG. 3 and the respective description in PCT/IB2017/052383 showing the design of an actuator which allows the rotation of a prism around one or two axes.

[0350] Rotation of the reflecting element around one or two axes moves the position of the camera FOV, wherein in each position a different portion a scene is captured in an image having the resolution of the digital camera. In this way a plurality of images of adjacent non-overlapping (or partially overlapping) camera FOV are captured and stitched together to form a stitched (also referred to as “composite”) image having an overall image area of an FOV greater than digital camera FOV.

[0351] In some examples the digital camera can be a folded Tele camera configured to provide a Tele image with a Tele image resolution, the folded Tele camera comprising a Tele image sensor and its Tele lens assembly is characterized with a Tele field of view (FOV.sub.T).

[0352] According to some examples, the folded Tele camera is integrated in a multiple aperture digital camera comprising at least one additional upright Wide camera configured to provide a Wide image with a Wide image resolution, being smaller than the Tele image resolution, the Wide camera comprising a Wide image sensor and a Wide lens module with a Wide field of view (FOV.sub.W); wherein FOV.sub.T is smaller than FOV.sub.W. Wherein rotation of the rotating reflecting element moves FOV.sub.T relative to FOV.sub.W.

[0353] The description of co-owned international patent applications PCT/IB2016/056060 and PCT/IB2016/057366 includes a Tele camera with an adjustable Tele field of view. As described in PCT/IB2016/056060 and PCT/IB2016/057366, rotation of the reflecting element around one or two axes moves the position of Tele FOV (FOV.sub.T) relative to the Wide FOV (FOV.sub.W), wherein in each position a different portion a scene (within FOV.sub.W) is captured in a “Tele image” with higher resolution. According to some examples, disclosed in PCT/IB2016/056060 and PCT/IB2016/057366, a plurality of Tele images of adjacent non-overlapping (or partially overlapping) Tele FOVs are captured and stitched together to form a stitched (also referred to as “composite”) Tele image having an overall image area of an FOV greater than FOV.sub.T. According to some examples, the stitched Tele image is fused with the Wide image generated by the Wide camera.

[0354] According to some examples, the digital camera further comprises or is otherwise operatively connected to a computer processing device, which is configured to control the operation of the digital camera (e.g. camera CPU). The digital camera can comprise a controller operatively connected to the actuator of the rotating reflecting element and configured to control its operation for rotating the rotating reflecting element.

[0355] The computer processing device can be responsive to a command requesting an image with a certain zoom factor and control the operation of the digital camera for providing images having the requested zoom. As mentioned in applications PCT/IB2016/056060 and PCT/IB2016/057366, in some examples a user interface (executed for example by the computer processing device) can be configured to allow input of user command being indicative of a requested zoom factor. The computer processing device can be configured to process the command and provide appropriate



instructions to the digital camera for capturing images having the requested zoom.

[0356] In some cases, if the requested zoom factor is a value between the FOV.sub.W and FOV.sub.T the computer processing device can be configured to cause the actuator of the reflecting element to move the reflecting element (by providing instruction to the controller of the actuator) such that a partial area of the scene corresponding to the requested zoom factor is scanned and a plurality of partially overlapping or non-overlapping Tele images, each having a Tele resolution and covering a portion of the partial area, are captured. The computer processing device can be further configured to stitch the plurality of captured imaged together in order to form a stitched image (composite image) having Tele resolution and an FOV greater than the FOV.sub.T of the digital camera. Optionally the stitched image can then be fused with the Wide image.

[0357] FIG. **14** is a schematic illustration of an example of a stitched image generated by capturing and stitching together 4 Tele images. In FIG. **14**, **1401** denotes FOV.sub.W, **1403** denotes FOV.sub.T at the center of FOV.sub.W and **1405** indicates the size of the requested zoom factor. In the illustrated example, four partially overlapping Tele-images (**1407**) are captured.

[0358] Notably, the overall area of the captured Tele-images (**1407**) is greater than the area of the requested zoom image (**1405**). The central part of the captured Tele-images is extracted (e.g. by the computer processing device as part of the generation of the stitched image) for generating the stitched image **1405**. This helps to reduce the effect of image artefacts resulting from transition from an image area covered by one image to an image area covered by a different image.

[0359] FIG. **15** is a schematic illustration of an example of a stitched image generated by capturing and stitching together 6 Tele images. FIG. **16** is a schematic illustration of an example of a stitched image generated by capturing and stitching together 9 Tele images. The same principles described with reference to FIG. **14** apply to FIGS. **15** and **16**. Notably, the output image resulting from the stitching can have a different width to height proportion than the single image proportion. For example, as illustrated in FIG. **15**, a single image can have 3:4 proportion and the output stitched image can have a 9:16 proportion.

[0360] It is noted that image stitching per se is well known in the art and therefore it is not explained further in detail.

[0361] FIG. **20** shows another exemplary embodiment of a lens module numbered **2000** which includes N lens elements L.sub.i (where “i” is an integer between 1 and N). In the example of FIG. **20**, N is equal to 5. For example, L.sub.1 is made of glass. The description above referring to lens module **200** holds also for lens module **2000**, with the necessary change of N from 4 to 5.

[0362] In some cases, both the first and last lens elements can be of increased dimensions in order to increase light impinging on the sensor. Examples of an optical lens module which is designed to deal with such a case is given in FIGS. **21-26**.

[0363] FIG. **21A-B** show another exemplary embodiment of a lens module numbered **2100** which includes N lens elements L.sub.i. In the example of FIG. **21A-B**, N is equal to 5. Lens module **2100** has the property of H.sub.L1=H.sub.LN. In FIGS. **21A-B**, lens module **2100** is shown without a lens barrel. FIG. **21A** shows light ray tracing of lens module **2100** while FIG. **21B** shows only the lens elements for more clarity. In addition, both figures show image sensor **202** and optical element **205**.

[0364] FIG. **22** shows schematically in a side view an exemplary lens module numbered **2200** for holding the lens elements of lens module **2100**. Lens module **2200** comprises a barrel **2202** having a cavity **2204** surrounded by walls **2206**. In lens module **2200**, a first subset of the lens elements is held inside the cavity, and a second subset of the lens elements is located externally to (outside) the barrel. Specifically, lens elements L.sub.2 to L.sub.N-1 are held within cavity **2204** and lens elements L.sub.1 and L.sub.N are located outside of barrel **2202** (i.e. lens elements L.sub.1 and L.sub.N are not within cavity **2204**). Lens element L.sub.1 and L.sub.N can be affixed (fixedly attached) to barrel **2202** by any appropriate mechanical link, such as an adhesive material.

[0365] In lens module **2200**, since lens elements L.sub.1 and L.sub.N are located outside of cavity

**2204**, the height  $H_{\text{sub.L1}}$  and  $H_{\text{sub.LN}}$  of, respectively, lens elements  $L_{\text{sub.1}}$  and  $L_{\text{sub.N}}$  can be substantially equal to the height of barrel **2202** (measured along the axis Y between external surfaces of opposite walls of barrel **2202**). Heights  $H_{\text{sub.L2}}$  to  $H_{\text{sub.LN-1}}$  of lens element  $L_{\text{sub.2}}$  to  $L_{\text{sub.N-1}}$  can be smaller than the height of barrel **2202**, marked with H. A numerical (non-limiting) example for lens module **2200** may have the following values:

$H_{\text{sub.L1}}=H_{\text{sub.L5}}=4$  mm,  $H_{\text{sub.L2}}=H_{\text{sub.L3}}=H_{\text{sub.L4}}=3.6$  mm.

[0366] FIG. **23** is a schematic representation of a side view of another optical lens module numbered **2300** for holding the lens elements of FIGS. **21A**, **21B**. Lens module **2300** comprises a barrel **2302** having a cavity **2304** surrounded by walls **2306**. In lens module **2300**, all the lens elements  $L_{\text{sub.1}}$  and  $L_{\text{sub.N}}$  are held (located) inside the cavity. Exemplarily, in lens module **2300**, a height  $H_{\text{sub.C}}$  of cavity **2304**, measured along an axis Y orthogonal to optical axis **103**, is variable along optical axis **103** (i.e. the Z axis). For each position along the Z axis, cavity height  $H_{\text{sub.C}}$  corresponds in this example to the distance between the internal parts **2308** of walls **2306** along axis Y. In other words,  $H_{\text{sub.C}}(Z)$  is not a constant function.

[0367] According to the example shown, cavity **2304** comprises a first portion **2310** in which first lens element  $L_{\text{sub.1}}$  is located, a second portion **2312** in which the other lens elements ( $L_{\text{sub.2}}$  to  $L_{\text{sub.N-1}}$ ) are located and a third portion **2314** in which last lens element  $L_{\text{sub.N}}$  is located. According to this example, heights  $H_{\text{sub.C}}(Z_{\text{sub.1}})$  of first portion **2310** and  $H_{\text{sub.C}}(Z_{\text{sub.3}})$  of third portion **2314** are greater than height  $H_{\text{sub.C}}(Z_{\text{sub.2}})$  of second (middle) portion **2312**. As a consequence, first lens element  $L_{\text{sub.1}}$  and last lens element  $L_{\text{sub.N}}$  (which are generally of greater dimensions, as mentioned above) are positioned respectively within first portion **2310** and third portion **2314** (respectively) of cavity **2304**, and at least some of the other lens elements are positioned within second portion **2312** of cavity **2304**.

[0368] According to this example, height  $H_{\text{sub.C}}(Z_{\text{sub.1}})$  of first portion **2310** is designed to fit with the height  $H_{\text{sub.L1}}$  of first lens element  $L_{\text{sub.1}}$ , height  $H_{\text{sub.C}}(Z_{\text{sub.2}})$  of second portion **2312** is designed to fit with height  $H_{\text{sub.L2}}$ ,  $H_{\text{sub.L3}}$  and  $H_{\text{sub.L4}}$  of lens elements  $L_{\text{sub.2}}$  to  $L_{\text{sub.4}}$  (in this example,  $H_{\text{sub.L2}}=H_{\text{sub.L3}}=H_{\text{sub.L4}}$ ) and height  $H_{\text{sub.C}}(Z_{\text{sub.3}})$  of third portion **2314** is designed to fit with the height  $H_{\text{sub.L5}}$  of last lens element  $L_{\text{sub.5}}$ .

[0369] The variable height of cavity **2304** along optical axis **103** can be obtained e.g. by using walls **2306** with a variable thickness. As shown in FIG. **23**, walls **2306** have a thinner thickness in first portion **2310** and third portion **2314** than in second portion **2312**. In other examples, walls **2306** may have a constant thickness but may have a stepped shape.

[0370] FIG. **24** is a schematic representation of a side view of another exemplary optical lens module numbered **2400** for holding the lens elements of FIGS. **21A**, **21B**. Lens module **2400** comprises a barrel **2402** having a cavity **2404** surrounded by walls **2406**. In this example, lens elements  $L_{\text{sub.2}}$  to  $L_{\text{sub.N-1}}$  are located within cavity **2404**. Lens elements  $L_{\text{sub.1}}$  and  $L_{\text{sub.N}}$  have a first part located inside cavity **2404** and a second part located outside of cavity **2404**; this is similar to lens element  $L_{\text{sub.1}}$  of FIG. **10A**. An edge **2408** of lens element  $L_{\text{sub.1}}$  and an edge **2410** of lens element  $L_{\text{sub.N}}$  has a stepped shape. Walls **2406** align edges **2408** and **2410** such that the center of lens elements  $L_{\text{sub.1}}$  and  $L_{\text{sub.N}}$  can be maintained in alignment with optical axis **103**.

[0371] Attention is now drawn to FIGS. **25A** to **25D**.

[0372] FIG. **25A** is a schematic representation of an exploded isometric view of another exemplary optical lens module numbered **2500** having a lens barrel **2502** and of a plurality of lens elements  $L_{\text{sub.1}}$  to  $L_{\text{sub.N}}$  (in this example  $N=4$ ) before their insertion into the barrel. FIG. **25B** depicts a cross-section view of lens module **2500** along plane Y-Z, FIG. **25C** depicts a cross-section view of lens module **2500** along plane X-Z, and FIG. **25D** depicts another isometric view of lens module **2500** after the insertion of the lens elements into the barrel.

[0373] Barrel **2502** comprises a cavity **2504** surrounded by walls **2506**. Lens elements  $L_{\text{sub.1}}$  to  $L_{\text{sub.N}}$  are located within cavity **2504**. Lens module **2500** may further include spacers  $R_{\text{sub.1}}$  to

R.sub.N-1. Each spacer R.sub.i is positioned between lens elements L.sub.i and L.sub.i+1. In some embodiments, one or more of spacers R.sub.1 to R.sub.N-1 may be used as an aperture stop(s). Spacers R.sub.1 to R.sub.N-1 can have an annular shape.

[0374] A height H of barrel **2502** is measured for example between external surfaces of opposite walls **2512** of barrel **2502** (e.g. along an axis Y orthogonal to optical axis **103**). In the examples of FIGS. **25A** to **25D**, a height H.sub.L1 of lens element L.sub.1 and a height H.sub.LN of lens element L.sub.N can be substantially equal to H or greater than H. Thus, lens elements L.sub.1 and L.sub.N can have a large height (therefore benefiting from an increased optical collection surface) while being located within the optical lens module which provides protection and mechanical support for lens elements L.sub.1 and L.sub.N. With this configuration, the center of lens elements L.sub.1 and L.sub.N can be maintained in alignment with optical axis **103**.

[0375] Similar to FIGS. **11A-11F** above, each lens element has an optical part and a mechanical part. According to some examples, the ratio between a height of the optical part (see H.sub.opt in FIG. **11F**) and the height of the lens element (see H.sub.L in FIG. **11F**) is greater for lens elements L.sub.1 and L.sub.N than for each of lens elements L.sub.2 to L.sub.N-1.

[0376] As shown in the figures, barrel **2502** can comprise slots **2510** on the top and bottom wall of barrel **2502** on its two endings: close to the object side and close to the image side. This allows lens elements L.sub.1 and/or L.sub.N to be substantially of the same height as the barrel, or to have a height which is greater than the barrel height, and to have a height which is greater than that of the other lens elements. In particular, lens elements L.sub.1 and/or L.sub.N can be tangent to slots **2510**, or at least parts of the lens elements L.sub.1 and/or L.sub.N can protrude through slots **2510**.

[0377] The structure of the lens as depicted in FIGS. **25A** to **25D** is thus also advantageous in terms of the manufacturing process, since lens elements L.sub.1 and L.sub.N can be of the height of the barrel (or can have a height which is greater than the height of the barrel) and can still be fastened to the internal surfaces of the walls of the barrel.

[0378] The assembly of lens module **2500** may be done using the following steps: [0379] 1.

Insertion of lens element L.sub.N from the image side of barrel **2502**. L.sub.N may be aligned to barrel **2502** due to the axial symmetry of both elements. [0380] 2. Fixedly attaching (e.g. gluing) L.sub.1 to barrel **2502**. [0381] 3. Insertion of other elements from the object side of barrel **2502** in the following order: R.sub.N-1, L.sub.N-1 . . . R.sub.2, L.sub.1. L.sub.1 to L.sub.N-1 and R.sub.1 to R.sub.N-1 may be aligned to barrel **2502** due to the axial symmetry of all elements. [0382] 4. Fixedly attaching (e.g. gluing) L.sub.N to barrel **2502**.

[0383] In one example, holes **2514** (FIG. **25D**) in barrel **2502** may be used to insert the glue to fasten lens elements L.sub.1 and L.sub.N in steps 2 and 4.

[0384] Attention is now drawn to FIGS. **26A** to **26C**. FIG. **26A** shows an isometric view of another exemplary lens module numbered **2600**. FIG. **26B** shows a side view of lens module **2600**. Lens module **2600** comprises a barrel **2602** having a cavity **2604**, and a plurality of lens elements L.sub.1 to L.sub.N. N is normally in the range of 3-7. In the non-limiting example of lens **2600**, N=4. Lens module **2600** has the first lens elements L.sub.1 and L.sub.N partially positioned or placed outside of barrel **2602**, while lens elements L.sub.2 to L.sub.N-1 are placed completely inside the barrel. L.sub.1 and L.sub.N are clearly seen in FIG. **26A**, while other lens elements are not seen in this view but can be seen in FIG. **26B**. As in previous examples, optical axis **103** serves as an axial symmetry axis for all lens elements L.sub.1 to L.sub.N. Each lens element L.sub.i has a height H.sub.Li defined along the Y axis. Lens elements L.sub.1 and L.sub.N may have a “stepped” shape, i.e. it has a front part with height H.sub.L1 (H.sub.LN) and a back part with height H.sub.L1B (H.sub.LNB), such that H.sub.L1>H.sub.L1B and H.sub.LN>H.sub.LNB. Lens module **2600** may further include spacers R.sub.1 to R.sub.N-1. Each spacer R.sub.i is positioned between lens elements L.sub.i and L.sub.i+1. In some embodiments, one or more of spacers R.sub.1 to R.sub.N-1 may be used as an aperture stop(s). In some embodiments some adjacent lens elements may not have a spacer therebetween.

[0385] Cavity **2604** may be made for example from opaque plastic and may be axial symmetric along optical axis **103**, like cavity **1720** in FIGS. **17A-17E**. In an exemplary embodiment, cavity **2604** may have a shape of cylinder as in embodiment **1700** (FIG. **17B**). In other exemplary embodiments, cavity **2604** may have other axial symmetric shapes, such as a cone, a series of cylinders.

[0386] FIG. **26C** shows a lens module **2601** which is similar to lens module **2600** with a single difference: barrel **2622** with cavity **2624** replaces barrel **2602** with cavity **2624**. Cavity **2624** has a shape of a series of cylinders in increasing sizes for each lens element; as can be seen in FIG. **17D**,  $H_{\text{sub.L1}} \leq H_{\text{sub.L2}} \leq H_{\text{sub.L3}} \leq H_{\text{sub.L4}} < H_{\text{sub.LNB}} < H_{\text{sub.L1}} = H_{\text{sub.LN}}$ . This feature may allow easier molding of barrel **2620** and/or easier assembly of lens elements  $L_{\text{sub.1}}$  to  $L_{\text{sub.4}}$  and spacers  $R_{\text{sub.1}}$  to  $R_{\text{sub.3}}$ . In other embodiments, the number of lens elements may differ from four, as mentioned above.

[0387] The assembly of lens module **2400** (or **2401**), and in particular the order of lens element insertion into the barrel, may be similar to the assembly steps of lens module **1700** above (FIGS. **17A-17D**).

[0388] Attention is now drawn to FIG. **27**. FIG. **27** shows an isometric view of a lens module **2700** which is similar to lens module **2500**, except that it has an added cover **2730** similar to cover **1830** of lens **1800**, with similar assembly steps.

[0389] Unless otherwise stated, the use of the expression “and/or” between the last two members of a list of options for selection indicates that a selection of one or more of the listed options is appropriate and may be made.

[0390] It should be understood that where the claims or specification refer to “a” or “an” element, such reference is not to be construed as there being only one of that element.

[0391] All patents and patent applications mentioned in this specification are herein incorporated in their entirety by reference into the specification, to the same extent as if each individual patent or patent application was specifically and individually indicated to be incorporated herein by reference. In addition, citation or identification of any reference in this application shall not be construed as an admission that such reference is available as prior art to the present disclosure.

## Claims

1. A mobile device, comprising: a first camera having a first field-of-view (FOV1) and a first image sensor; and a second camera having a second field-of-view (FOV2), a second effective focal length (EFL2) and a second total track length (TTL2), an optical lens module including  $N$  lens elements  $L_{\text{sub.1}}-L_{\text{sub.N}}$ , a first reflecting element, and a second image sensor, wherein  $\text{FOV2} < \text{FOV1}$ , wherein the first image sensor is configured to receive a first light from a first direction through a first optical path between an object and the first image sensor, wherein the second image sensor is configured to receive a second light from the first direction through a second optical path between the object and the second image sensor, wherein the first reflecting element is disposed in the second optical path between the optical lens module and the second image sensor, wherein  $\text{EFL2}/\text{TTL2} \leq 1.1$ , wherein each lens element is denoted  $L_{\text{sub.i}}$  where  $1 \leq i \leq N$ , wherein each lens element comprises a respective front surface  $S_{\text{sub.2i-1}}$  and a respective rear surface  $S_{\text{sub.2i}}$ , the lens element surfaces marked  $S_{\text{sub.k}}$  where  $1 \leq k \leq 2N$ , wherein each lens element surface has a clear aperture value  $\text{CA}(S_{\text{sub.k}})$ , and wherein a clear aperture value  $\text{CA}(S_{\text{sub.1}})$  of surface  $S_{\text{sub.1}}$  is greater than a clear aperture value of each of surfaces  $S_{\text{sub.2}}$  to  $S_{\text{sub.2N}}$ .
2. The mobile device of claim 1, wherein  $3 \leq N \leq 7$ .
3. The mobile device of claim 1, wherein the  $N$  lens elements are arranged along an optical axis such that lens element  $L_{\text{sub.N}}$  is a last lens element facing the second image sensor.
4. The mobile device of claim 1, further comprising a second reflecting element in the second optical path.

5. The mobile device of claim 1, wherein the second image sensor has a sensor diagonal SD, and wherein a ratio  $CA(S.sub.2N)/SD$  between a clear aperture value  $CA(S.sub.2N)$  of surface  $S.sub.2N$  and SD is smaller than 0.8.
6. The mobile device of claim 5, wherein  $CA(S.sub.2N)/SD < 0.7$ .
7. The mobile device of claim 5, wherein  $CA(S.sub.2N)/SD < 0.65$ .
8. The mobile device of claim 1, wherein  $1.04 \leq CA(S.sub.1)/CA(S.sub.2) \leq 1.23$ .
9. The mobile device of claim 1, wherein  $0.59 \leq CA(S.sub.2N)/CA(S.sub.1) \leq 0.76$ .
10. The mobile device of claim 1, wherein the second camera has an effective focal length EFL in the range of 8 mm to 16 mm.
11. The mobile device of claim 1, wherein the second camera has an effective focal length EFL in the range of 11 mm to 16 mm.
12. The mobile device of claim 1, wherein the second camera has a F-number  $\leq 3$ .
13. The mobile device of claim 1, wherein the second camera has a F-number  $\leq 2.76$ .
14. The mobile device of claim 1, wherein L1 has a positive refractive power.
15. The mobile device of claim 1, wherein the second camera has a second back focal length (BFL2), and wherein  $BFL2 \geq 0.3 \times TTL2$ .
16. The mobile device of claim 1, wherein  $EFL2/TTL2 \leq 1$ .
17. The mobile device of claim 1, wherein rotation of the first reflecting element around one or two axes moves the position of FOV2.
18. The mobile device of claim 1, wherein one or more of the lens elements are cut lens elements.
19. The mobile device of claim 2, wherein  $N=5$ .
20. The mobile device of claim 1, wherein the mobile device is a smartphone.
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